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Person support apparatus with adjustable exit detection zones

Abstract

A person support apparatus, such as a bed, cot, stretcher, or the like, for supporting an occupant includes an exit detection system that has, in some embodiments, multiple user-selectable zones. An alert is triggered when the occupant crosses a boundary of the selected zone. A first zone has a boundary that is adjusted based on a condition of the person support apparatus, such as the width of the support deck or other characteristic. A second zone has a boundary that is not adjusted based on the condition of the person support apparatus, but which may have a boundary that is dependent on an initial location of the occupant when the exit detection system is armed. The initial location of the occupant may correspond to a center of gravity of the occupant.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
RE28754	12/1975	Cook et al.	N/A	N/A
4175263	12/1978	Triplett et al.	N/A	N/A
4242672	12/1979	Gault	N/A	N/A
5802640	12/1997	Ferrand et al.	N/A	N/A
5844488	12/1997	Musick	N/A	N/A
6009570	12/1999	Hargest et al.	N/A	N/A
6049281	12/1999	Osterweil	N/A	N/A
6067019	12/1999	Scott	N/A	N/A
6208250	12/2000	Dixon et al.	N/A	N/A
6585645	12/2002	Hutchinson	N/A	N/A
6754516	12/2003	Mannheimer	N/A	N/A
6791460	12/2003	Dixon et al.	N/A	N/A
7253366	12/2006	Bhai	N/A	N/A
7304580	12/2006	Sullivan et al.	N/A	N/A
7369680	12/2007	Trajkovic et al.	N/A	N/A
7437787	12/2007	Bhai	N/A	N/A
7502498	12/2008	Wen et al.	N/A	N/A
7538659	12/2008	Ulrich et al.	N/A	N/A
7541934	12/2008	Fredriksson et al.	N/A	N/A
7612666	12/2008	Badawy	N/A	N/A
7629890	12/2008	Sullivan et al.	N/A	N/A
7656299	12/2009	Gentry et al.	N/A	N/A
7666151	12/2009	Sullivan et al.	N/A	N/A
7714728	12/2009	Koblasz	N/A	N/A
7834770	12/2009	Kazuno	N/A	N/A
7987069	12/2010	Rodgers et al.	N/A	N/A
8042206	12/2010	Genaro	N/A	N/A

8082160	12/2010	Collins, Jr. et al.	N/A	N/A
8258963	12/2011	Dixon et al.	N/A	N/A
8266742	12/2011	Andrienko	N/A	N/A
8272087	12/2011	Westermann	N/A	N/A
8281433	12/2011	Riley et al.	N/A	N/A
8284047	12/2011	Collins, Jr. et al.	N/A	N/A
8381336	12/2012	Kazuno et al.	N/A	N/A
8525680	12/2012	Riley et al.	N/A	N/A
8723677	12/2013	Kiani	N/A	N/A
8756078	12/2013	Collins, Jr. et al.	N/A	N/A
8821418	12/2013	Meger et al.	N/A	N/A
8830070	12/2013	Dixon et al.	N/A	N/A
8838411	12/2013	Kazuno et al.	N/A	N/A
8844073	12/2013	Riley et al.	N/A	N/A
8902066	12/2013	Parker et al.	N/A	N/A
8907287	12/2013	Vanderpohl	N/A	N/A
9013313	12/2014	Paine	N/A	N/A
9044367	12/2014	Yakam et al.	N/A	N/A
9138173	12/2014	Penninger et al.	N/A	N/A
9177465	12/2014	Vanderpohl, III	N/A	N/A
9179863	12/2014	Brauers et al.	N/A	N/A
9198815	12/2014	Murai	N/A	N/A
9226696	12/2015	Kiani	N/A	N/A
9253891	12/2015	Williams	N/A	N/A
9295600	12/2015	Receveur	N/A	N/A
9320444	12/2015	Hayes et al.	N/A	N/A
9378632	12/2015	Venetianer et al.	N/A	N/A
9445751	12/2015	Young et al.	N/A	N/A
9489818	12/2015	Vanderpohl, III	N/A	N/A
9504619	12/2015	Murai	N/A	N/A
9539156	12/2016	Lemire et al.	N/A	N/A
9549675	12/2016	Riley et al.	N/A	N/A
9579047	12/2016	Clark et al.	N/A	N/A
9727061	12/2016	Liu et al.	N/A	N/A
9750654	12/2016	Brondum	N/A	N/A
9754476	12/2016	Lemire et al.	N/A	N/A
9775758	12/2016	Riley et al.	N/A	N/A
9782108	12/2016	Shimizu	N/A	N/A
9795321	12/2016	Shimizu	N/A	N/A
9814410	12/2016	Kostic et al.	N/A	N/A
9861321	12/2017	Collins, Jr. et al.	N/A	N/A
9875633	12/2017	Pirio et al.	N/A	N/A
9883809	12/2017	Klap et al.	N/A	N/A
10357185	12/2018	Kostic	N/A	A61B 5/6892
10617327	12/2019	Kostic	N/A	A61G 7/0524
10786408	12/2019	Sidhu	N/A	A61G 7/018

11490834	12/2021	Sukumaran	N/A	G08B 29/185
11800995	12/2022	Sukumaran	N/A	A61G 7/0509
12036161	12/2023	Connell	N/A	A61G 7/0524
12144607	12/2023	Kostic	N/A	A61G 7/018
12268642	12/2024	Sidhu	N/A	A61B 5/1115
2003/0018241	12/2002	Mannheimer	N/A	N/A
2007/0038155	12/2006	Kelly, Jr. et al.	N/A	N/A
2009/0044334	12/2008	Parsell et al.	N/A	N/A
2011/0068932	12/2010	Flocard et al.	N/A	N/A
2011/0153915	12/2010	Zitlaw	N/A	N/A
2012/0075464	12/2011	Derenne et al.	N/A	N/A
2014/0115784	12/2013	Johannigman et al.	N/A	N/A
2014/0313030	12/2013	Ten Kate et al.	N/A	N/A
2014/0343889	12/2013	Ben Shalom et al.	N/A	N/A
2015/0108443	12/2014	Huh et al.	N/A	N/A
2015/0238123	12/2014	Yakam et al.	N/A	N/A
2015/0323388	12/2014	Kostic et al.	N/A	N/A
2015/0327777	12/2014	Kostic et al.	N/A	N/A
2016/0022218	12/2015	Hayes et al.	N/A	N/A
2016/0023144	12/2015	Fitzgerald et al.	N/A	N/A
2016/0106345	12/2015	Kostic et al.	N/A	N/A
2016/0128610	12/2015	Kostic et al.	N/A	N/A
2016/0140307	12/2015	Brosnan et al.	N/A	N/A
2016/0140827	12/2015	Derenne et al.	N/A	N/A
2016/0193095	12/2015	Roussy et al.	N/A	N/A
2017/0042750	12/2016	Murai	N/A	N/A
2017/0098359	12/2016	Sidhu et al.	N/A	N/A
2017/0143566	12/2016	Elku et al.	N/A	N/A
2017/0156638	12/2016	Ribble et al.	N/A	N/A
2017/0224253	12/2016	Berlin et al.	N/A	N/A
2017/0243459	12/2016	Sidhu	N/A	A61G 7/005
2018/0108239	12/2017	Pirio et al.	N/A	N/A
2020/0214599	12/2019	Kostic	N/A	A61G 7/0524
2021/0007919	12/2020	Sidhu	N/A	A61G 7/005
2021/0045950	12/2020	Connell	N/A	A61G 7/015
2021/0052197	12/2020	Sukumaran	N/A	A61B 5/6892
2021/0338505	12/2020	Nahavandi	N/A	A61B 5/747

2021/0353179	12/2020	Sukumaran	N/A	A61G 7/0528
2023/0067526	12/2022	Sukumaran	N/A	A61G 7/0509
2023/0233102	12/2022	Paul	5/611	A61B 5/6892
2024/0277543	12/2023	Sukumaran	N/A	A61G 7/005
2024/0299226	12/2023	Sukumaran	N/A	A61B 5/1115
2025/0031999	12/2024	Kostic	N/A	A61B 5/1115

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2020204399	12/2020	AU	A61B 5/1115
3085085	12/2020	CA	A61B 5/1115
3171373	12/2020	CA	A61B 5/1115
3181260	12/2021	CA	A61B 5/004
101558011	12/2008	CN	N/A
1976433	12/2010	EP	N/A
2725507	12/2012	EP	N/A
2013240601	12/2012	JP	N/A
9730351	12/1996	WO	N/A
9834577	12/1997	WO	N/A
2009029996	12/2008	WO	N/A
2009055635	12/2008	WO	N/A
2011113070	12/2010	WO	N/A
2015136424	12/2014	WO	N/A
2015167643	12/2014	WO	N/A
WO-2020102129	12/2019	WO	A61B 5/1115
WO-2021231674	12/2020	WO	A61B 5/1115
WO-2022072340	12/2021	WO	A61B 5/0004
WO-2023064182	12/2022	WO	A61B 5/115
WO-2023064183	12/2022	WO	A61G 7/005
WO-2023064184	12/2022	WO	N/A
WO-2023225090	12/2022	WO	N/A
WO-2023225246	12/2022	WO	A61B 5/1115

OTHER PUBLICATIONS

Canadian Office Action for Canadian Application No. Application No. 3,085,085, dated Oct. 1, 2024. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. provisional patent applications 62/889,254 filed Aug. 20, 2019, by inventors Sujay Sukumaran et al. and entitled PERSON SUPPORT APPARATUS WITH ADJUSTABLE EXIT DETECTION ZONES, and to U.S. provisional patent application Ser. No. 63/024,066 filed May 13, 2020, by inventors Grady Sertic et al. and entitled PATIENT SUPPORT APPARATUS WITH AUTOMATIC EXIT DETECTION MODES OF OPERATION, the complete disclosures of both of which are incorporated herein by reference.

BACKGROUND

(1) The present disclosure relates to person support apparatuses, such as beds, cots, stretchers, recliners, or the like. More specifically, the present disclosure relates to person support apparatuses that include sensors for monitoring the motion and/or activity of an occupant of the person support apparatus and issuing an alert if the occupant is about to, or does, exit the person support apparatus. (2) Existing hospital beds and/or stretchers often include an exit detection system that is adapted to detect when a patient has exited the bed, or when a patient may be about to exit the bed. Typically, such beds include circuitry for providing an audio or visual alert when such an exit or pre-exit situation is detected. In many cases, the bed or stretchers include circuitry for transmitting a signal to a remote location, such as a nurses' station, so that the appropriate caregivers are notified of the exit, or pre-exit condition, and can respond appropriately. The exit detection system itself may be implemented in a variety of manners, including using a plurality of force sensors.

SUMMARY

(3) According to various embodiments, an improved person support apparatus is provided that adjusts a size, shape, position, or other characteristic of one or more alert zones of an exit detection system based on one or more conditions of the person support apparatus. The exit detection system issues an alert if an occupant of the person support apparatus moves outside of whichever one of the alert zones has been designated as the active alert zone. The exit detection system may also include an arming zone for each of the alerting zones. The exit detection system may adjust the boundary of one or more of the arming zones in addition to adjusting the boundary of one or more of the alerting zones. The adjustable alerting zones allow improved alerting regarding an occupant's intention to exit the person support apparatus, including, but not limited to, advance notification of such an exit and a reduction in false alarms. The improved monitoring of the occupant's motion may also provide a reduction in failures to arm the exit detection system.

(4) A person support apparatus according to one embodiment of the present disclosure includes a litter frame, a plurality of lifts adapted to raise and lower the litter frame, a support deck, and an exit detection system. The support deck is supported on the litter frame and configured to have an adjustable width. The exit detection system is adapted to issue an alert if an occupant of the person

support apparatus moves outside of a boundary of an alert zone. The exit detection system comprises a plurality of force sensors adapted to output signals corresponding to downward forces exerted on the support deck, and a controller. The controller is in communication with the force sensors and is configured to determine a width of the support deck and adjust the boundary of the alert zone based on the deck width.

(5) A person support apparatus according to another embodiment of the present disclosure includes a litter frame, a plurality of lifts adapted to raise and lower the litter frame, a support deck, an arming control, and an exit detection system. The support deck is supported on the litter frame and configured to have an adjustable width. The exit detection system is adapted to issue an alert if an occupant of the person support apparatus moves outside of an alert zone, and the exit detection system includes an arming zone having a boundary in which the occupant must remain present during an arming process of the exit detection system. The exit detection system comprises a plurality of force sensors adapted to output signals corresponding to downward forces exerted on the support deck, and a controller. The controller communicates with the force sensors and is configured to determine a condition of the person support apparatus; to adjust the boundary of the arming zone based on the condition of the person support apparatus; to arm the exit detection system in response to a user activating the arming control if the occupant remains in the arming zone during the arming process of the exit detection system; and to not arm the exit detection system in response to the user activating the arming control if the occupant does not remain in the arming zone during the arming process of the exit detection system.

(6) A person support apparatus according to another embodiment of the present disclosure includes a litter frame, a plurality of lifts adapted to raise and lower the litter frame, a support deck, and an exit detection system. The exit detection system is adapted to issue an alert if an occupant of the person support apparatus moves outside of a user-selected one of a plurality of alert zones. A first one of the plurality of alert zones has a variable boundary defined independently of a location of the occupant when the exit detection system is initially armed, and a second one of the plurality of alert zones has a static boundary defined by a location of the occupant when the exit detection system is initially armed. The exit detection system comprises a plurality of force sensors adapted to output signals corresponding to downward forces exerted on the support deck, and a controller. The controller communicates with the plurality of force sensors and is configured to determine a condition of the person support apparatus. The controller is further configured, after the exit detection system has been armed, to adjust the variable boundary of the first alert zone, but not the static boundary of the second alert zone, based on the condition of the person support apparatus.

(7) A person support apparatus according to another embodiment of the present disclosure includes a litter frame, a plurality of lifts adapted to raise and lower the litter frame, a support deck, and an exit detection system. The exit detection system is adapted to issue an alert if an occupant of the person support apparatus moves outside of a user-selected one of a plurality of alert zones. The plurality of alert zones includes a first alert zone and a second alert zone. The exit detection system comprises a plurality of force sensors adapted to output signals corresponding to downward forces exerted on the support deck, and a controller. The controller communicates with the plurality of force sensors and is configured to determine a condition of the person support apparatus, to adjust the boundary of the first alert zone based on the condition of the person support apparatus, and to not adjust the boundary of the second alert zone based on the condition of the person support apparatus.

(8) A person support apparatus according to another embodiment of the present disclosure includes a litter frame, a plurality of lifts adapted to raise and lower the litter frame, a support deck, a mattress interface, and an exit detection system. The mattress interface is adapted to receive status data from a mattress configured to provide lateral rotation therapy to an occupant of the person support apparatus. The lateral rotation therapy comprises rotating the occupant laterally in a direction. The exit detection system is adapted to issue an alert if an occupant of the person support

apparatus moves outside of a boundary of an alert zone. The exit detection system comprises a plurality of force sensors adapted to output signals corresponding to downward forces exerted on the support deck, and a controller. The controller communicates with the mattress interface and the plurality of force sensors. The controller is configured to receive the status data from the mattress interface, to determine a status of the lateral rotation therapy based on the status data; and to adjust the boundary based on the status of the lateral rotation therapy.

(9) A person support apparatus according to another embodiment of the present disclosure includes a litter frame, a plurality of lifts adapted to raise and lower the litter frame, a support deck, and an exit detection system. The exit detection system is adapted to issue an alert if an occupant of the person support apparatus moves outside of a boundary of an alert zone. The exit detection system comprises a plurality of force sensors adapted to output signals corresponding to downward forces exerted on the support deck, and a controller. The controller communicates with the plurality of force sensors and is configured to determine an incline angle of the litter frame and to adjust the boundary of the alert zone based on the incline angle.

(10) A person support apparatus according to another embodiment of the present disclosure includes a litter frame, a plurality of lifts adapted to raise and lower the litter frame, a support deck, and an exit detection system. The exit detection system is adapted to issue an alert if an occupant of the person support apparatus moves outside of a boundary of an alert zone. The exit detection system comprises a plurality of force sensors adapted to output signals corresponding to downward forces exerted on the support deck, and a controller. The controller communicates with the plurality of force sensors and is configured to detect an addition of a non-occupant object to the support deck or a removal of a non-occupant object from the support deck, and to adjust the boundary based on the addition or removal of the non-occupant object.

(11) Before the various embodiments disclosed herein are explained in detail, it is to be understood that the claims are not to be limited to the details of operation or to the details of construction and the arrangement of the components set forth in the following description or illustrated in the drawings. The embodiments described herein are capable of being practiced or being carried out in alternative ways not expressly disclosed herein. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including” and “comprising” and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items and equivalents thereof. Further, enumeration may be used in the description of various embodiments. Unless otherwise expressly stated, the use of enumeration should not be construed as limiting the claims to any specific order or number of components. Nor should the use of enumeration be construed as excluding from the scope of the claims any additional steps or components that might be combined with or into the enumerated steps or components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a person support apparatus according to one embodiment of the disclosure;

(2) FIG. 2 is a perspective view of a portion of a base of the person support apparatus of FIG. 1, including a plurality of force sensors of an exit detection system.

(3) FIG. 3 is a perspective view of an adjustable width support deck of the person support apparatus of FIG. 1.

(4) FIG. 4 is a perspective view of an expandable seat section of the adjustable width support deck of FIG. 3 showing elements for expanding and latching the seat section.

(5) FIG. 5 is a block diagram of a portion of a control system of the person support apparatus of

FIG. 1, including an exit detection system.

(6) FIG. 6 is a diagram illustrating one manner in which the exit detection system of FIG. 5 determines if an occupant is about to exit or not.

(7) FIG. 7 is a plan view diagram of a reference frame for the person support apparatus of FIG. 1 showing multiple user-selectable zones for the exit detection system.

(8) FIG. 8 is a plan view of a portion of a user interface for the person support apparatus of FIG. 1, the user interface displaying an arbitrary screen and including a control for arming the exit detection system.

(9) FIG. 9 is a schematic view of an arming screen configured to be displayed on the user interface of FIG. 8 during arming of the exit detection system.

(10) FIG. 10 is a schematic view of a failure screen configured to be displayed on the user interface of FIG. 8 if the exit detection system fails to arm.

(11) FIG. 11 is a schematic view of a control screen for the exit detection system configured to be displayed on the user interface of FIG. 8.

(12) FIG. 12 is a plan view diagram of the person support apparatus illustrating an alert zone used to trigger an exit alarm and an arming zone used to arm the exit detection system, shown with the support deck in a first width.

(13) FIG. 13 is a plan view diagram of the person support apparatus of FIG. 12 showing a modification of the zones in response to the support deck expanding to a second width greater than the first width of FIG. 12.

(14) FIG. 14 is a plan view diagram of the person support apparatus of FIG. 12 showing another modification of the zones in response to the support deck expanding to a third width greater than the second width of FIG. 13.

(15) FIG. 15 is a plan view diagram of the person support apparatus illustrating multiple user-selectable alert zones and associated arming zones, shown with the support deck in a first width.

(16) FIG. 16 is a plan view diagram of the person support apparatus of FIG. 15 showing a modification of multiple zones in response to the support deck expanding to a second width.

(17) FIG. 17 is a plan view diagram of the person support apparatus of FIG. 15 showing another modification of multiple zones in response to the support deck expanding to a third width.

(18) FIG. 18 is a diagram of the person support apparatus illustrating an alert zone and an arming zone within a reference frame, shown with a litter frame in a horizontal orientation.

(19) FIG. 19 is a diagram of the person support apparatus of FIG. 18 showing a modification of the zones in response to the litter frame moving to a Trendelenburg orientation.

(20) FIG. 20 is a diagram of the person support apparatus of FIG. 18 showing another modification of the zones in response to the litter frame moving to a reverse Trendelenburg orientation.

(21) FIG. 21 is a diagram of the person support apparatus of FIG. 18 showing a modification of multiple user-selectable zones in response to the litter frame moving to a Trendelenburg orientation.

(22) FIG. 22 is a diagram of the person support apparatus of FIG. 18 showing another modification of the multiple user-selectable zones in response to the litter frame moving to a reverse Trendelenburg orientation.

(23) FIG. 23 is a transverse cross-section of a mattress for the person support apparatus of FIG. 1, the mattress configured to provide lateral rotation therapy to an occupant.

(24) FIG. 24 is a view similar to FIG. 23, illustrating lateral rotation of the occupant to the left by a leftward incline of the mattress.

(25) FIG. 25 is a view similar to FIG. 23, illustrating lateral rotation of the occupant to the right by a rightward incline of the mattress.

(26) FIG. 26 is a plan view diagram of the person support apparatus illustrating an alert zone used to trigger an exit alarm and an arming zone used to arm the exit detection system, when the mattress is in a non-rotated or horizontal orientation as shown in FIG. 23.

- (27) FIG. 27 is a plan view diagram of the person support apparatus of FIG. 26 showing a modification of the zones in response to lateral rotation to a leftward incline as shown in FIG. 24.
- (28) FIG. 28 is a plan view diagram of the person support apparatus of FIG. 26 showing another modification of the zones in response to lateral rotation to a rightward incline as shown in FIG. 25.
- (29) FIG. 29 is a schematic view of a lateral rotation therapy control screen configured to be displayed on the user interface of FIG. 8.
- (30) FIG. 30 is a plan view diagram of the person support apparatus illustrating an alert zone used to trigger an exit alarm and an arming zone used to arm the exit detection system, shown with no non-occupant object on the support deck.
- (31) FIG. 31 is a plan view diagram of the person support apparatus of FIG. 30 showing a modification of the zones in response to a non-occupant object being added to the support deck.
- (32) FIG. 32 is a plan view diagram of the person support apparatus of FIG. 30 showing the modified alert and arming zones after the non-occupant object is added.
- (33) FIG. 33 is a plan view diagram of the person support apparatus illustrating an alert zone used to trigger an exit alarm and an arming zone used to arm the exit detection system, shown with a non-occupant object on the support deck.
- (34) FIG. 34 is a plan view diagram of the person support apparatus of FIG. 33 showing a modification of the zones in response to the non-occupant object being removed from the support deck.
- (35) FIG. 35 is a plan view diagram of the person support apparatus of FIG. 33 showing the modified alert and arming zones after the non-occupant object is removed.
- (36) FIG. 36 is a plan view diagram of the person support apparatus of FIG. 33 showing another modification of the alert and arming zones after a non-occupant object is added.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (37) An illustrative person support apparatus 20 that may incorporate one or more aspects of the present disclosure is shown in FIG. 1. Although the particular form of person support apparatus 20 illustrated in FIG. 1 is a bed adapted for use in a hospital or other medical setting, it will be understood that person support apparatus 20 could, in different embodiments, be a cot, a stretcher, a gurney, a recliner, a residential bed, or any other structure capable of supporting a person, whether stationary or mobile and/or whether medical or residential.
- (38) In general, person support apparatus 20 includes a base 22 having a plurality of wheels 24, a pair of lifts 26 supported on the base, a litter frame 28 supported on the lifts 26, and a support deck 30 supported on the litter frame 28. Person support apparatus 20 further includes a headboard 32, a footboard 34, and a plurality of siderails 36. Siderails 36 are all shown in a raised position in FIG. 1 but are each individually movable to a lower position in which ingress into, and egress out of, person support apparatus 20 is not obstructed by the lowered siderails 36. In some embodiments, siderails 36 may be moved to one or more intermediate positions as well.
- (39) Lifts 26 are adapted to raise and lower litter frame 28 with respect to base 22. Lifts 26 may be hydraulic actuators, electric actuators, or any other suitable device for raising and lowering litter frame 28 with respect to base 22. In the illustrated embodiment, lifts 26 are operable independently so that the tilting of litter frame 28 with respect to base 22 can also be adjusted, to place the litter frame 16 in a flat or horizontal orientation, a Trendelenburg orientation, or a reverse Trendelenburg orientation. That is, litter frame 28 includes a head end 38 and a foot end 40, each of whose height can be independently adjusted by the nearest lift 26. Person support apparatus 20 is designed so that when an occupant lies thereon, his or her head will be positioned adjacent head end 38 and his or her feet will be positioned adjacent foot end 40.
- (40) Litter frame 28 provides a structure for supporting support deck 30, the headboard 32, footboard 34, and siderails 36. Support deck 30 provides a support surface for a mattress (not shown in FIG. 1), or other soft cushion, so that a person may lie and/or sit thereon. The top surface of the mattress or other cushion forms a support surface for the occupant.

(41) Support deck **30** is made of a plurality of sections, some of which are pivotable about generally horizontal pivot axes. In the embodiment shown in FIG. **1**, support deck **30** includes a head section **42**, a seat section **44**, a thigh section **46**, and a foot section **48**. Head section **42**, which is also sometimes referred to as a Fowler section, is pivotable about a generally horizontal pivot axis between a generally horizontal orientation (shown in FIG. **1**) and a plurality of raised positions (not shown). Thigh section **46** and foot section **48** may also be pivotable about generally horizontal pivot axes.

(42) FIG. **2** illustrates in greater detail a portion of the base **22**, including two of the wheels **24**. Base **22** includes a wheel frame **50** and a base frame **52** supported by the wheel frame **50**. A portion of the wheel frame **50** is shown transparently in FIG. **2** to better illustrate details of the base **22**. The wheel frame **50** supports the plurality of wheels **24** proximate the corners of the base frame **52**. A brake pedal **54** may permit locking one or more of the wheels **24** in a full stop state or a steered state, in addition to an unlocked state that permits the wheels **24** to both swivel and rotate.

(43) The lower ends of the lifts **26** are connected to the base frame **52**. Through this connection, the base frame **52** supports litter frame **28**, support deck **30**, footboard **34**, the headboard, and siderails **36**, and other components of the person support apparatus **10**.

(44) Base frame **52** includes a plurality of force sensors **56** resting on the wheel frame **50**. In the illustrated embodiment, the force sensors **56** are proximate the corners of the base frame **52**, with two of four force sensors **56** shown in FIG. **2**. The force sensors **56** may be load cells, or other types of force sensors, such as, but not limited to, linear variable displacement transducers and/or any one or more capacitive, inductive, and/or resistive transducers that are configured to produce a changing output in response to changes in the force exerted against them. The force sensor **56** may be electronically connected to control circuitry through an electrical contact (not shown).

(45) Although the illustrated embodiment of person support apparatus **20** includes a total of four force sensors **56**, two of which are shown in FIG. **2**, it will be understood by those skilled in the art that different numbers of force sensors **56** may be used in accordance with the principles of the present disclosure. Force sensors **56** are configured to support base frame **52**. More specifically, force sensors **56** are configured such that they provide complete and exclusive mechanical support for base frame **52** and all of the components that are supported on base frame **52** (e.g., lifts **26**, litter frame **28**, support deck **30**, footboard **34**, the headboard, siderails **36**, etc.). Because of this construction, force sensors **56** are adapted to detect the weight of not only those components of person support apparatus **20** that are supported by the base frame **52** (including base frame **52** itself), but also any objects or persons who are wholly or partially being supported by support deck **30**. By knowing the weight of the components of the person support apparatus **20**, or by taring the scale before the patient enters the person support apparatus **20**, a measurement of the weight of the patient may be obtained from the load cells. The outputs of force sensors **56** are part of an exit detection system **58** described in greater detail below.

(46) In some alternative embodiments, person support apparatus **20** is constructed with load cells **56** positioned in locations other than base frame **52**. For example, in at least one alternative embodiment, person support apparatus **20** is constructed with a litter frame and lift construction of the type disclosed in commonly assigned U.S. patent application Ser. No. 15/266,575 filed Sep. 15, 2016 by inventors Anuj K. Sidhu et al. and entitled PERSON SUPPORT APPARATUSES WITH EXIT DETECTION SYSTEMS, the complete disclosure of which is incorporated herein by reference. When person support apparatus **20** is constructed with the lift and litter frame construction of the type disclosed in this '575 application, the load cells **56** may be positioned on the top of the lifts such that the entire weight of the litter frame is supported on the load cells, as illustrated more clearly, for example, in FIGS. 2 and 3 of the aforementioned '575 application. Still other manners of positioning the load cells **56** within the person support apparatus **20**, and/or other locations for the load cells **56**, may be utilized.

(47) Person support apparatus **20** comprises features for extending the width of its support deck **30**

to accommodate patients of varying sizes. The width may be adjusted in any increments, for example between a first or minimum width, a second or intermediate width, and a third or expanded/maximum width. In one embodiment, person support apparatus **20** is configured to allow the support deck **30** to be adjusted to a first width of thirty-six inches, a second width of forty-two inches, and a third width of forty-eight inches, although these numerical values may, of course, take on different values in different embodiments. Further, support deck **30** may be configured to be adjustable to more than three different widths, in some embodiments, or less than three different widths in other embodiments.

(48) In addition to being adjustable in width, person support apparatus **20** is adjustable in height via the lifts **26** adapted to raise and lower litter frame **28** with respect to base **22**, as disclosed above. In further embodiments, the person support apparatus **20** may be adjustable in length, with the length between the head end **38** and foot end **40** being adjustable to accommodate patients of varying height.

(49) As used herein, the term “longitudinal” refers to a direction parallel to an axis between the head end **38** and the foot end **40**, where a head-to-foot distance is parallel to a longitudinal axis and is referred to as the length of the person support apparatus **20**. The terms “transverse” or “lateral” refer to a direction perpendicular to the longitudinal direction and parallel to a surface on which the person support apparatus **20** rests, where a side-to-side distance is parallel to the transverse or lateral axis and is referred to as the width of the person support apparatus **20**.

(50) Referring to FIGS. **3-4**, the width adjustable support deck **30** may comprise multiple expandable sections **60**, or extension pans, which can be adjusted laterally by one or more length extendable actuators **64**. The length extendable actuators **64** may be operated (extended or retracted) independently or together, and may be operated manually or automatically. Manually adjusting the width may be accomplished by pulling or pushing the support deck **30**, optionally using handles **62** coupled to the sections **60**, in a direction lateral to a longitudinal axis of the person support apparatus **20**, the longitudinal axis extending between the head end **38** and the foot end **40**. The support deck **30** may comprise any mechanism configured to permit manually adjusting the width of the support deck **30**, for example a rack and pinion mechanism.

(51) The head section **42**, seat section **44**, and thigh section **46** (and in some embodiments, foot section **48**) comprise one or more pairs of expandable sections **60** which are positioned under the exposed top surface of deck section **42-46** (and **48**, in some embodiments) when the support deck **30** is at its narrowest width. In one example shown in FIG. **4**, taking the seat section **44** as an example, two deck sections **60** are linked by an extendible actuator **64**, with the extension of the actuator **64** driving the linked deck sections **60** laterally in opposite directions to provide a wider surface. When the support deck **30** is expanded, the extension sections **60** that otherwise reside under the deck sections are exposed to provide an extended surface on which a larger mattress may rest. The litter frame **28** does expand with the support deck **30**, and in some embodiments, the foot section **48** is not expandable.

(52) In some embodiments, adjustable width deck **30** is constructed in any of the manners disclosed in commonly assigned U.S. patent application Ser. No. 62/887,977 filed Aug. 16, 2019, by inventors Jason Connell et al. and entitled PATIENT SUPPORT WITH DECK WIDTH MONITORING AND CONTROL, the complete disclosure of which is incorporated herein by reference. Other types of constructions of adjustable width deck **30** can, of course, be used.

(53) It will also be understood by those skilled in the art that the non-deck portions of person support apparatus **20** can also or alternatively be designed with other types of mechanical constructions, such as, but not limited to, that described in commonly assigned, U.S. Pat. No. 10,130,536 to Roussy et al., entitled PATIENT SUPPORT USABLE WITH BARIATRIC PATIENTS, the complete disclosure of which is incorporated herein by reference. The mechanical construction of the non-deck portions of person support apparatus **20** may also take on forms different from what is disclosed in the aforementioned reference.

(54) As shown in FIG. 5, person support apparatus **20** includes an exit detection system **58** that is adapted to determine when an occupant, such as, but not limited to, a patient, of person support apparatus **20** is moving and is likely to exit person support apparatus **20**. More specifically, exit detection system **58** is adapted to determine when an occupant is leaving, or is likely to leave, and to issue an alert and/or notification to appropriate personnel so that proper steps can be taken in response to the occupant's departure, or imminent departure, in a timely fashion. The particular structural details of exit detection system **58** can vary widely. It will be understood by those skilled in the art that components of exit detection system **58** may be added or omitted from one or more of the embodiments of exit detection system **58** that are discussed herein.

(55) In the embodiment shown in FIG. 5, exit detection system **58** includes force sensors **56** and a controller **76**. Exit detection system **58** is in communication with one or more apparatus condition sensors **78** that detect a condition of person support apparatus **20**, one or more occupant condition sensors **80** that detect a condition of an occupant of the person support apparatus **20**, and a user interface **82**. The condition sensors **78**, **80** generate one or more output signals received by the controller **76**. Based on the output signals received, the controller **76** can make an adjustment to an alert zone and/or an arming zone of the exit detection system **58**, as will be discussed in greater detail below. Force sensors **56** may also be used to detect one or more conditions of the occupant, as the force sensors **56** may be used to detect a weight or center of gravity of an occupant of the person support apparatus **20**, as described in further detail below. When forces sensors **56** are used to detect one or more of the occupant's conditions, a separate occupant condition sensor **80** may be omitted from exit detection system **58**.

(56) The condition sensors **78**, **80** may take on any of a variety of different forms, including one or more load cells, pressure sensors such as piezoelectric and piezoresistive sensors, Hall Effect sensors, capacitive sensors, resonant sensors, thermal sensors, limit switches, gyroscopes, accelerometers, motion sensors, ultrasonic sensors, range sensors, potentiometers, magnetostrictive sensors, electrical current sensors, voltage detectors, and/or any other suitable types of sensors for carrying out their associated functions. Regardless of the specific form, the condition sensors **78**, **80** report outputs to controller **76** and controller **76** uses the output, in at least some embodiments, to adjust an alert zone and/or an arming zone of the exit detection system **58**.

(57) Some non-limiting examples of condition sensors **78** include one or more width sensors **78a** adapted to detect a deck width of the support deck **30**, one or more angle sensors **78b** adapted to detect an incline of the litter frame **28**, or one or more position sensors **78c** adapted to detect the position of one of the plurality of lifts **26**. One or more additional apparatus condition sensors may be provided, such as a plurality of siderail sensors adapted to detect the position of each of the siderails **36**. One manner of using siderail sensors to make an adjustment to an alert zone of an exit detection system is disclosed in commonly assigned U.S. Patent Application Publication No. 2017/0098359 to Sidhu et al., entitled PERSON SUPPORT APPARATUS WITH EXIT DETECTION SYSTEMS, the complete disclosure of which is incorporated herein by reference.

(58) Width sensors **78a** output signals that indicate a current width of the support deck **30**. This width refers to a transverse or lateral dimension of the support deck **30** in a direction perpendicular to the longitudinal direction of person support apparatus **20**. In some embodiments, width sensors **78a** are coupled to each individual expandable section **60** such that the width of each section **60** is determined. In other embodiments, a single width sensors **78a** may be utilized that collectively senses whether all of sections **60** are at a particular width or not. Still other variations of width sensors **78a** may also or additionally be used.

(59) Angle sensors **78b** output signals that indicate a current incline angle of the litter frame **28**, and can include tilt sensors or potentiometers for sensing an angular orientation of the litter frame **28**, or any other suitable sensor for detecting the angular orientation of the litter frame **28**.

(60) Position sensors **78c** are adapted to detect the position of one or more movable components of person support apparatus **20**. In some embodiments, position sensors **78c** are adapted to detect how

far each lift **26** has extended. From this, controller **76** can determine an incline angle of the litter frame **28**. In other embodiments, position sensors **78c** can determine an angle of head section **42** (the Fowler section) relative to horizontal (or relative to one or more references on person support apparatus **20** itself). In still other embodiments, multiple types of position sensors **78c** are included so that, for example, position sensors **78c** report the current incline angle of the litter frame **28**, height of litter frame **28**, and head section angle to controller **76**. In some embodiments, as will be discussed more below, controller **76** uses the position information from the one or more position sensors **78c** as a factor in adjusting one or more zones of the exit detection system **58** and/or determining if the occupant is about to exit from person support apparatus **20**.

(61) Some non-limiting examples of condition sensors **80** include a vital sign sensor adapted to detect one or more vital signs of the occupant of person support apparatus **20**. Force sensors **56** may also be used to detect one or more vital signs of the occupant. Manners for detecting the occupant's vital signs using force sensors **56** are disclosed in commonly assigned U.S. Pat. No. 7,699,784 to Wan Fong et al., entitled SYSTEM AND METHOD FOR MONITORING VITAL SIGNS, the complete disclosure of which is hereby incorporated herein by reference. One manner of using vital sign sensors to make an adjustment to an alert zone of an exit detection system is disclosed in commonly assigned U.S. Patent Application Publication No. 2017/0098359 to Sidhu et al., incorporated above.

(62) Force sensors **56** are adapted to detect downward forces exerted by an occupant of support deck **30**. Thus, when an occupant is positioned on support deck **30** and substantially still (i.e. not moving in a manner involving accelerations that cause forces to be exerted against support deck **30**), force sensors **56** will detect the weight of the occupant (as well as the weight of any components of person support apparatus **20** that are supported—directly or indirectly—by force sensors **56**). Force sensors **56** are also used to determine a center of gravity of the occupant, as will be discussed in greater detail below, in order to determine if the occupant is about to exit person support apparatus **20**. In alternative embodiments, the outputs from force sensors **56** are analyzed, not to determine a center of gravity, but instead to determine a weight distribution and/or a change in weight distribution, such as by determining one or more ratios of the relative weights sensed by the force sensors **56** and using them to determine if the occupant is about to exit person support apparatus **20**. In still other embodiments, force sensors **56** may be modified to detect forces other than, or in addition to, the downward forces exerted by the occupant. In yet other embodiments, the outputs from force sensors **56** are analyzed to determine if a non-occupant object has been added to or removed from the person support apparatus **20**. Other types of sensors may also or alternatively be used for determining the occupant's weight.

(63) Exit detection system **58** may also include one or more auxiliary inputs **84**. Auxiliary inputs **84** are constructed as ports into which one or more sensors, cables, or other devices are coupled. The outputs from the sensors, cables, or other devices, are communicated to controller **76** and used as a factor, in some embodiments, in adjusting an alert zone and/or an arming zone of the exit detection system **58**. The number of auxiliary inputs **84** may vary. Indeed, in some embodiments, exit detection system **58** includes no auxiliary inputs **84**. However, when one or more auxiliary inputs **84** are included, such auxiliary inputs **84** may be configured in any of the following manners: as a wired port for coupling to a cable (e.g. an Ethernet port for coupling to an Ethernet cable, a USB port for coupling to a USB cable, a Controller Area Network (CAN) transceiver for coupling exit detection system **58** to a CAN bus, another type of embedded network port, etc.), as a port for coupling to a wireless transceiver (e.g. a Wi-Fi transceiver, a Bluetooth transceiver, a ZigBee transceiver, a near field communication (NFC) transceiver, etc.), as a port for coupling to one or more additional sensors, and/or as a port for coupling to other devices.

(64) When coupled to an Ethernet cable or a Wi-Fi transceiver, one or more auxiliary inputs **84** may be used to communicate with, and receive information from, a healthcare facility local area network (LAN). More specifically, inputs **84** may receive information from an Electronic Medical

Record (EMR) system that is in communication with the LAN. Such information may include any one or more of the following types of information about the occupant of person support apparatus **20**: his or her gender, his or her height and/or weight, his or her fall risk assessment, and/or other information about the occupant. Exit detection system **58** optionally uses one or more of these items of information, in some embodiments, as factors in adjusting an alert zone and/or an arming zone of the exit detection system **58**. One manner of using gender, height, weight, a fall risk assessment, and other information about the occupant to make an adjustment to an alert zone of an exit detection system is disclosed in commonly assigned U.S. Patent Application Publication No. 2017/0098359 to Sidhu et al., incorporated above. In some alternative embodiments, exit detection system **58** does not couple directly to the healthcare facility LAN, but instead communicates with one or more other components onboard person support apparatus **20** that are in communication with the LAN.

(65) Controller **76** is constructed of any electrical component, or group of electrical components, that are capable of carrying out the functions described herein. In many embodiments, controller **76** is a conventional microcontroller, although not all such embodiments need include a microcontroller. In general, controller **76** includes any one or more microprocessors, microcontrollers, field programmable gate arrays, systems on a chip, volatile or nonvolatile memory, discrete circuitry, and/or other hardware, software, or firmware that is capable of carrying out the functions described herein, as would be known to one of ordinary skill in the art. Such components can be physically configured in any suitable manner, such as by mounting them to one or more circuit boards, or arranging them in other manners, whether combined into a single unit or distributed across multiple units. The instructions followed by controller **76** in carrying out the functions described herein, as well as the data necessary for carrying out these functions, are stored in a memory (not labeled) accessible to controller **76**.

(66) User interface **82** communicates with controller **76** and enables a user of person support apparatus **20** to control one or more aspects of person support apparatus **20**, including exit detection system **58**. User interface **82** is implemented in the embodiment shown in FIG. **1** as a control panel having a plurality of controls. The controls—which may be buttons, dials, switches, or other devices—allows a user to control various aspects of exit detection system **58**, such as, but not limited to, selecting a mode of operation of exit detection system **58** and/or arming and disarming exit detection system **58**. User interface **82** may also include a display **86** for displaying information regarding exit detection system **58**. Display **86** may be a touchscreen that displays one or more controls and/or one or more of the control screens discussed below. Display **86** may comprise an LED display, OLED display, or another type of display.

(67) Although FIG. **1** illustrates user interface **82** mounted to footboard **34**, it will be understood that user interface **82** can be positioned elsewhere, and/or that one or more additional user interfaces can be added to person support apparatus **20** in different locations, such as the siderails **36**, for controlling various aspects of exit detection system **58**. In addition, one or more user interfaces may be communicatively coupled to person support apparatus **20** but physically positioned remote from person support apparatus **20**, such as, but not limited to, a computer tablet, a smart phone, a computer station, etc.

(68) Controller **76** of exit detection system **58** is adapted to determine the center of gravity of the occupant using the outputs from force sensors **56**. Controller **76** uses this center of gravity to determine whether or not the occupant is about to exit from person support apparatus **20**. In one embodiment, exit detection system **58** determines this center of gravity using the system and method disclosed in commonly assigned U.S. Pat. No. 5,276,432 to Travis, entitled PATIENT EXIT DETECTION MECHANISM FOR HOSPITAL BED, the complete disclosure of which is incorporated herein by reference. In other embodiments, other algorithms may be used.

(69) As shown more clearly in FIG. **6** controller **76** determines the center of gravity of the occupant **O** in a planar coordinate frame of reference, such as reference frame **88**. Reference frame **88**

includes an X-axis **90** and a Y-axis **92**. X-axis **90** is generally parallel to the footboard **34** of person support apparatus **20** while Y-axis **92** is generally parallel to a siderail **36**, two of which are depicted in a lowered position in FIG. **6**. Other coordinate systems can be used. Regardless of which coordinate system is used, controller **76** knows the location of force sensors **56** in the particular coordinate system that is used. In the example shown in FIG. **6**, force sensors **56** are shown in known locations **94**.

(70) In the illustrative example shown in FIG. **6**, controller **76** has determined the occupant's center of gravity to be at a location **96**. Controller **76** compares this center of gravity **96** to the active alert zone **98** (explained in further detail below) that is defined in reference frame **88** and determines whether the center of gravity **96** is inside or outside of this alert zone **98**. If center of gravity **96** moves outside of the alert zone **98**, controller **76** issues an alert indicating that the occupant is about to exit from person support apparatus **20**. When determining whether the center of gravity **96** is outside or inside of the alert zone **98**, controller **76** may first compute the center of gravity in a first one of the directions of coordinate frame of reference **88** (X direction or Y direction), compare that value to the corresponding boundaries of the zone in that particular direction and, if it is inside the boundaries, compute the center of gravity in the other direction of coordinate frame of reference **88** (X direction or Y direction).

(71) As shown in FIG. **6**, there are three different alert zones **98a**, **b**, and **c**. Alert zones **98a**, **b**, and **c** have different sizes, allowing the occupant to engage in different amounts of movement prior to triggering an exit alert. A user selects which one of the alert zones **98a-c** will be the active alert zone using user interface **82**. Controller **76** then repetitively recalculates the occupant's center of gravity **96** based upon the outputs from force sensors **56** and compares the calculated center of gravity **96** to the active zone. If the center of gravity **96** is within the active alert zone **98**, no exit alert is issued. If the center of gravity **96** moves outside of the active zone **98**, controller **76** issues an alert. In some embodiments, in order to avoid issuing an alert based upon transient weight signals shifting the center of gravity **96** outside of the active zone **98** for a fleeting moment, controller **76** only issues an alert if the center of gravity **96** moves outside of the active zone **98** for more than a threshold amount of time (which may be on the order of seconds or a fraction of a second).

(72) In some embodiments, controller **76** of exit detection system **58** is adapted to execute an arming process when it is initially armed and/or when it is already armed and a user switches the active zone from one zone to another. During the arming process, controller monitors movement of the occupant using the outputs from force sensors **56** and determines a center of gravity of the occupant. If the occupant moves too much, or moves outside of a specified region referred to herein as an arming zone, during the arming process, controller **76** does not arm the exit detection system **58** and notifies the user that the arming process failed. If, on the other hand, the occupant's movement levels remain below a threshold and the occupant's position remains within the arming zone, controller **76** completes the arming process and notifies the user that the exit detection system has been successfully armed. In the illustrated embodiment, there is a separate arming zone associated with each alerting zone.

(73) As shown more clearly in FIG. **7**, controller **76** compares the center of gravity **96** of the occupant to the active one of arming zones **100** (i.e. the one corresponding to the alert zone selected by the user or caregiver) and determines whether the center of gravity **96** is inside or outside of this active zone **100** and if the occupant is settled or moving. If center of gravity **96** is outside of the active zone **100** (discussed below), controller **76** does not arm the exit detection system **58** and issues a notification indicating that the exit detection system **58** is unable to arm. When determining whether the center of gravity **96** is outside or inside of the active arming zone **100**, controller **76** may first compute the center of gravity in a first one of the directions of coordinate frame of reference **88** (X direction or Y direction), compare that value to the corresponding boundaries of the zone in that particular direction and, if it is inside the boundaries,

compute the center of gravity in the other direction of coordinate frame of reference **88** (X direction or Y direction). Alternatively, controller **76** may compute the center of gravity in both the X and Y directions and then compare the calculated center of gravity to the boundary defined by the arming zone **100**.

(74) If center of gravity **96** is inside of the active arming zone **100** (discussed below) but the occupant is not settled, controller **76** does not arm the exit detection system **58** and issues a notification indicating that the exit detection system **58** is unable to arm. When determining whether the occupant is settled or not settled, controller **76** may analyze output signals from the force sensors **56** to determine if the outputs change at a rate greater than a threshold speed for more than a threshold time (for example, on the order of 2 seconds). Once controller **76** determines the occupant is settled, controller **76** will arm the exit detection system **58**. In some embodiments, in order to mitigate arming failures, if the occupant is not settled initially, controller **76** will continue to analyze the output signals for a longer period of time, or timeout period (for example, on the order of 15 seconds). If the occupant does not settle within the timeout period, controller **76** does not arm the exit detection system **58** and issues a notification indicating that the exit detection system **58** is unable to arm.

(75) As shown in FIG. 7, there are three different arming zones **100a**, **b**, and **c**. Arming zones **100a-c** have different sizes, and are each associated with one of the three alert zones **98a-c** of FIG. 6. In the example shown herein, alert zone **98a** is associated with arming zone **100a**, alert zone **98b** is associated with arming zone **100b**, and alert zone **98c** is associated with arming zone **100c**. Upon user-selection of one of the alert zones **98a-c** to be the active alert zone using user interface **82**, the associated arming zone **100a-c** is automatically selected by controller **76** as the active arming zone.

(76) Each zone **98**, **100** has a boundary **102**, **104** defining the area covered by the zone **98**, **100**, respectively. In the embodiment shown herein, the boundaries **102**, **104** are indicated by dashed lines. When the exit detection system **58** is armed and an occupant of person support apparatus **20** moves such that his or her center of gravity **96** travels outside of the active alert zone **98**, i.e. crosses the boundary **102**, controller **76** issues an alert. To arm the exit detection system **58**, the center of gravity **96** must be within the boundary **104** of the active arming zone **100** and the occupant settled. In still other embodiments, exit detection system **58** includes more zones **98**, **100** than the three shown in FIG. 7, while in other embodiments, exit detection system **58** includes fewer zones than the three shown in FIG. 7 (including, in some embodiments, only a single alert zone **98** and arming zone **100**).

(77) Each arming zone **100a-c** is smaller in size than its associated alert zone **98a-c**, and has its boundary contained within a boundary of the associated alert zone **98a-c**. Although FIGS. 6-7 illustrate each alert zone **98a-c** and arming zone **100a-c** as having a generally rectangular shape, it will be understood that the alert zones **98a-c** and arming zones **100a-c** do not all have to have the same shape. Further, it will also be understood that any one or more of the zones can be shaped in other manners besides rectangles. In some embodiments, any one or more of the zones **98**, **100** are squares, parallelograms, other quadrilaterals, circles, ovals, or any combination of arcs, straight lines, curves, and/or other shapes.

(78) As noted, user interface **82** includes an arming control that enables a user to turn exit detection system **58** on and off (arm and disarm exit detection system **58**), as well as allowing a user to select different modes which are used for arming the system and triggering an exit alert. FIG. 8 depicts a portion of the user interface **82** (FIG. 1) including the display **86**. Display **86** is configured to display a plurality of different screens thereon, one of which (screen **106**) is shown in FIG. 9. The user interface **82** includes controls for accessing all of the functionality of person support apparatus **20**, including the exit detection system **58**. For example, user interface **82** includes an arming control **108** for arming exit detection system **58**. Upon user-activation of the arming control **108**, controller **76** is operable to arm exit detection system **58**. The arming control **108** can optionally show the status of the exit detection system **58**, such as by illuminating in green when the exit

detection system **58** is armed.

(79) While the exit detection system **58** attempts to arm, an arming screen **110**, such as that shown in FIG. **9**, is displayed on the display **86** of the user interface **82**. From this screen **110**, the caregiver can cancel the arming attempt by selecting a cancel control **112**, or can return to a home screen by selecting a home control **114**. While waiting for the occupant to settle, the arming screen **110** can optionally show a message **116** indicating that the occupant must settle before the system will arm (i.e., “Movement Detected; Keep patient still”).

(80) If the exit detection system **58** fails to arm, a failure screen **118** shown in FIG. **10** is displayed on the display **86** of the user interface **82**. The failure screen **118** includes a message **120** notifying the caregiver that the exit detection system **58** has failed to arm, as well as a reason for the failure to arm (i.e., “Movement on the bed has prevented to Bed Alarm from arming”). From this screen **118**, the caregiver can turn off the failure notification and return to the home screen by selecting a turn off control **122**.

(81) If the exit detection system **58** successfully arms, or if the user cancels the arming attempt by selecting cancel control **112** (FIG. **9**), a control screen **124** shown in FIG. **11** is displayed on the display **86** of the user interface **82**. From this control screen **124**, the caregiver can select an alert mode using mode controls **126**, **128**, **130**, can select to turn off or disarm the exit detection system **58** by selecting a turn off control **132**, or can return to the home screen by selecting a home control **134**. It is noted that if the exit detection system **58** is already armed, selecting the arming control **108** on the user interface **82** will also display control screen **124**.

(82) The mode of exit detection system **58** can be set to low, medium, or high using mode controls **126**, **128**, **130**. The modes correspond to the alert zones **98a-c** of person support apparatus **20** that are monitored by exit detection system **58**. By selecting one of the modes, the user selects which one of the alert zones **98a-c** will be the active alert zone, and the associated arming zone **100a-c** is automatically selected as the active arming zone. For example, with the “high” mode control **130** selected, exit detection system **58** monitors the first, smallest zone **98a** at or near the center of the support deck **30**. With the “medium” mode control **128** selected, exit detection system **58** monitors the second zone **98b** at or near the center of the support deck **30** which is larger than the first zone **98a**. With the “low” mode control **126** selected, exit detection system **58** monitors the third zone **98c** at or near the center of the support deck **30**, which is larger than both the first and second zones **98a**, **98b**. When first armed for a new patient, the default state for the exit detection system **58** is the medium mode.

(83) Returning to FIG. **7**, the boundary **102,104** of at least one of the zones **98**, **100** is dynamic. That is, at least one of the zones **98**, **100** has a size, shape, and/or location that varies based upon one or more criteria that will be discussed more below. In some embodiments, all of the boundaries **102**, **104** are dynamic.

(84) Controller **76** can adjust the boundary **102,104** of at least one of the dynamic zones **98**, **100** by changing a shape of the boundary **102,104**, changing a dimension of the boundary **102,104**, changing an area defined within the boundary **102,104**, changing a location of the boundary **102,104**, or any combination thereof. Changing a shape of the boundary **102,104** may include moving at least one portion of the boundary **102,104** laterally, longitudinally, or a combination thereof, and/or changing between various polygonal shapes, including rectangular, square, parallelogram, other quadrilaterals, circle, oval, or any combination of arcs, straight lines, curves, and/or other shapes. Changing a dimension of the boundary **102,104** may include increasing or decreasing a width or length of the zone **98**, **100**. Changing an area defined within the boundary **102,104** may include increasing or decreasing the size of the zone **98**, **100**. Changing a location of the boundary **102,104** may include shifting the boundary **102**, **104** while optionally maintaining the same shape or area for the zone **98**, **100**.

(85) Some of the criteria used by controller **76** to vary the boundary **102**, **104** of one or more zones **98**, **100** include the following: (a) the width of the support deck **30** of person support apparatus **20**;

(b) an incline angle of the litter frame **28**; (c) the status of a lateral rotation therapy; and (d) the addition or removal of a non-occupant object. Some additional the factors or criteria used by controller **76** to vary the boundary **102**, **104** of one or more zones **98**, **100** include the following: (a) the weight of the occupant; (b) the height of the occupant; (c) a ratio of the occupant's height and weight; (d) the gender of the occupant; (e) a fall risk assessment of the occupant; (f) values of one or more vital signs of the occupant; (g) a position of the siderails **36**; (h) a downward force being applied to one or more of the siderails **36**; (i) a position or orientation of one or more other components of person support apparatus **20**; (j) the environment or surroundings in which person support apparatus **20** is positioned; (k) a proximity of a caregiver to person support apparatus **20**; (l) a time of day; (m) one or more medical conditions of the occupant; (n) a body orientation of the occupant; (o) a current height of litter frame **28**, (p) movement of the occupant (e.g. sitting up), and/or other factors. It will be understood that controller **76** is programmed in some embodiments to use only a single one of these factors, while in other embodiments controller **76** is programmed to use any two or more of these of the factors in any possible combination.

(86) In some embodiments of exit detection system **58**, the user of person support apparatus **20**, such as the caregiver, can use user interface **82** to select the individual factors that are to be used by exit detection system **58**, or to select one or more predetermined combinations of factors that are to be used by exit detection system **58** when determining whether to issue an exit alert or not. In other embodiments, the particular combination of factors used by exit detection system **58** is preprogrammed. In still other embodiments, exit detection system **58** includes one or more user-selectable modes in which one or more preprogrammed combinations of factors are used by controller **76** to determine if an exit alert should be issued, but also includes one or more user-selectable modes that allow the user to customize the factors used by controller **76** when determining if an exit alert should be issued or not.

(87) In at least one embodiment in which controller **76** is configured to adjust the boundary **102** of an alert zone **98** based on the height of litter frame **28**, controller **76** receives signals from lifts **26** (and/or from the motor controller that oversees operation of lifts **26**) and uses those signals to determine a current height of litter frame **28**. In this embodiment, controller **76** contracts the size of the currently active alert zone **98** when the higher litter frame **28** is raised. Similarly, when litter frame **28** is lowered, controller **76** expands the size of the currently active alert zone **98**. Controller **76** applies this height-based expansion and contraction only to the medium and large alert zones **98b** and **98c**, and keeps small alert zone **98a** at a constant size, regardless of litter height.

(88) FIGS. **12-14** illustrate one manner in which controller **76** adjusts the boundary **102**, **104** of the zones **98**, **100** in response to the width of the support deck **30** being changed. FIG. **12** illustrates the person support apparatus **20** with support deck **30** at a first or minimum width LW1, with alert zone **98** defined by boundary **102** and arming zone **100** defined by boundary **104**. When an occupant of person support apparatus **20** moves such that his or her center of gravity **96** travels outside of alert zone **98**, i.e. crosses the boundary **102**, controller **76** issues an alert. When the center of gravity **96** of the occupant is inside of arming zone **100**, i.e. within the boundary **104**, controller **76** may arm the exit detection system **58**.

(89) As can be seen in FIG. **13**, when the width of support deck **30** is increased to a second or intermediate width LW2, controller **76** increases the size of the alert zone **98** by moving the boundary **102** outward along at least one side of the alert zone **98** to increase the area covered by the alert zone **98**. This allows the occupant's center of gravity **96** to approach the sides of the person support apparatus **20** more closely than it otherwise could (if the alert zone **98** remained the same as shown in FIG. **12**) without triggering an exit alert. Controller **76** also increases the size of the arming zone **100** by moving the boundary **104** outward along at least one side of the arming zone **100** to increase the area covered by the arming zone **100**. This allows the occupant's center of gravity **96** to be within a wider area of the person support apparatus **20** than it otherwise could (if the arming zone **100** remained the same as shown in FIG. **12**) while still permitting arming of the

exit detection system **58**.

(90) As can be seen in FIG. **14**, when the width of the support deck **30** is increased to a third or maximum width **LW3**, controller **76** increases the size of the alert zone **98** by moving the boundary **102** outward along at least one side of the alert zone **98** to increase the area covered by the alert zone **98**. This allows the occupant's center of gravity **96** to approach the sides of the person support apparatus **20** more closely than it otherwise could (if the alert zone **98** remained the same as shown in FIG. **12** or in FIG. **13**) without triggering an exit alert. Controller **76** also increases the size of the arming zone **100** by moving the boundary **104** outward along at least one side of the arming zone **100** to increase the area covered by the arming zone **100**. This allows the occupant's center of gravity **96** to be within a wider area of the person support apparatus **20** than it otherwise could (if the arming zone **100** remained the same as shown in FIG. **12** or in FIG. **13**) while still permitting arming of the exit detection system **58**.

(91) With respect to the width of support deck **30**, exit detection system **58** receives this information either directly from width sensors **78a** or it is input into exit detection system **58**. When input, it may be input via a caregiver entering the width information via user interface **82**, or it may be input by communicating with length extendable actuators **64**, **68**. Regardless of the source of the width data, controller **76** uses this information to adjust one or more of the zones **98**, **100**. That is, controller **76** uses the width to adjust one or more of the boundaries **102**, **104** of one or more of the zones **98**, **100** based upon this width.

(92) In some embodiments, the adjustments to the width of the zone **98**, **100** made in response to changes in the deck width are made incrementally based upon one or more predefined ranges of width. For example, a support deck width falling anywhere within a first range can cause controller **76** to adjust the width of the zones **98**, **100** by a first amount, support deck widths falling anywhere within a second range can cause controller **76** to adjust the width of the zones **98**, **100** by a second amount, etc. In some embodiments, only two ranges are used, while in other embodiments, other numbers of ranges are used.

(93) Controller **76** is also programmed in some embodiments to change the height of one or more zones **98**, **100** based upon the width of person support apparatus **20**. In other embodiments, controller **76** only changes the width of one or more zones **98**, **100**, or changes both the height and width based upon the width of person support apparatus **20**. The height of a zone refers to the length of the zone from a point on its boundary closest to head end **38** to a point on its boundary closest to foot end **40**.

(94) Although FIGS. **12-14** illustrate only a single alert zone **98** and a single arming zone **100**, it will be understood that, in some embodiments, controller **76** changes the size, shape, and/or location of multiple zones in response to the changing width. Further, it will be understood that the adjustment in size of the zones **98**, **100** illustrated in FIGS. **12-14** are merely one example of the types of zone adjustments that may be implemented by controller **76** in response to the changing width. Other types of zone adjustments are also possible, including an adjustment in shape and/or location.

(95) In some embodiments, controller **76** is configured to only expand the width of an alert zone **98** in response to an expansion of the deck **30** if all of the sections of the deck **30** are expanded. In such embodiments, if the user only expands, say, the width of seat section **44** from its minimum width to its intermediate width and the width of the other sections of support deck **30** (e.g. sections **42**, **46**, and **48**) remain at their minimum width, controller **76** keeps zone **100** at its narrowest width (e.g. the width shown in FIG. **12**). Similarly, controller **76** is configured in some embodiments to automatically reduce the width of the alert zone **98** if any one of the sections of support deck **30** are contracted. Thus, for example, if all of the deck sections are at their maximum width (e.g. FIG. **14**) and a user contracts head section **42** to its intermediate width, controller **76** automatically contracts the alert zone **98** to its intermediate size (e.g. what is shown in FIG. **13**). Further, if the user contracts head section **42** to its narrowest width (while the other sections of support deck **30** remain

at their widest width), controller **76** automatically contracts alert zone **98** to its narrowest size (e.g. what is shown in FIG. **12**).

(96) In an alternative embodiment, controller **76** is configured to adjust the alert zone **98** in response to each section of deck **30** being expanded or contracted. Thus, for example, if only seat section **44** is expanded, controller **76** is configured to make an adjustment to zone **98** that takes into account the lone expansion of seat section **44**. In this embodiment, if, after seat section **44** is expanded, the user subsequently expands, say, head section **42** to the same width as the seat section **44**, controller **76** is configured to make another adjustment to zone **98** that takes into account this expansion of head section **42**. In this fashion, controller **76** is configured to make piecemeal adjustments to the alert zone **98** based upon each individual section of support deck **30** that is widened or narrowed.

(97) FIGS. **15-17** illustrate one manner in which controller **76** adjusts the boundary **102**, **104** of multiple user-selectable zones in response to the deck width being changed. As can be seen, when the width of deck **30** is increased to the second or intermediate width **LW2** or to the third or maximum width **LW3**, controller **76** increases the size of multiple alert zones **98b**, **98c** by moving their boundaries **102b**, **102c** outward along at least one side of the zones **98b**, **98c** to increase the area covered by the zones **98b**, **98c**. Controller **76** also increases the size of multiple arming zones **100b**, **100c** by moving their boundaries **104b**, **104c** outward along at least one side of the arming zones **100b**, **100c** to increase the area covered by the arming zones **100b**, **100c**.

(98) In at least one embodiment, at least one of the multiple user-selectable zones is not adjusted in response to the deck width being changed. Specifically, controller **76** adjusts the boundaries **102b**, **102c**, **104b**, **104c** of those zone(s) **98b**, **98c**, **100b**, **100c** that have a dynamic boundary. Those zone(s) that have a static boundary, i.e. zones **98a**, **100a**, do not have their boundaries adjusted. As can be seen in FIGS. **15-17**, zones **98a** and **100a** correspond to the smallest alert zone **98a** and arming zone **100a**, which correspond to the “high” alert mode selectable by the caregiver in FIG. **11**.

(99) In one embodiment, while zones **98a**, **100a** are static with respect to changes in the width of the support deck **30**, the initial location of zones **98a**, **100a** is dynamic with respect to the patient's center of gravity at the time exit detection system **58** is initially armed, or at the time the user switches the active zone of an already armed exit detection system **58** from one of the other zones **98b** or **98c** to zone **98a**. That is, zones **98a** and **100a** are centered at a location corresponding to the patient's center of gravity at the time exit detection system **58** is armed with zone **98a** as the active zone, or at the time the user selects zone **98** as the active zone. Once defined at this location, zones **98a** and **100a** do not change their size, shape, or location in response to changes in the width of support deck **30**. In fact, in some embodiments, controller **76** does not thereafter adjust zones **98a** and **100a** for any reason other than if exit detection system **58** is disarmed and later re-armed, or the user selects another zone **98b** or **98c** to be the active zone and thereafter re-selects zone **98a** again as the active zone.

(100) In the aforementioned embodiment, controller **76** is configured to not change the initial location of zones **98b** or **98c** (or **100b** or **100c**) based on the patient's center of gravity at the time exit detection system **58** is armed. That is, regardless of where the patient is currently located when one of zones **98b** or **98c** is initially armed, or changed to be the current active zone, controller **76** defines the location of these zones in a manner that is fixed with respect to geometry of person support apparatus **20**. As a result, controller **76** is configured, in at least one embodiment, to initially define (at the moment of arming or activation) the location of zones **98b** and **98c** in a manner independent of the patient's position at that time, and to subsequently adjust zones **98b** and **98c** based on changes to the width of support deck **30**. In that same embodiment, controller **76** is configured to initially define (at the moment of arming or activation) the location of zone **98a** in a manner that is dependent of the patient's position at that time, and to subsequently not adjust the size, shape, or location of zone **98a** in response to changes to the width of support deck **30**.

(101) FIGS. **18-20** illustrate one manner in which controller **76** adjusts the boundary **102**, **104** of the zones **98**, **100** in response to the incline angle of litter frame **28** of the person support apparatus **20** being changed. As shown in the figures, the litter frame **28** can be tilted or inclined relative to the base **22** by controlling the height of lifts **26**. In FIG. **18**, the incline angle is zero since the litter frame **28** is oriented horizontally. Using reference frame **88**, alert zone **98** is defined by boundary **102** and arming zone **100** is defined by boundary **104**. When an occupant of person support apparatus **20** moves such that his or her center of gravity **96** travels outside of alert zone **98**, i.e. crosses the boundary **102**, controller **76** issues an alert. When the center of gravity **96** of the occupant remains inside of arming zone **100** during the arming process of exit detection system **58**, controller **76** is adapted to allow the exit detection system **58** to be armed.

(102) As can be seen in FIG. **19**, when the litter frame **28** is inclined to elevate the foot end **40** above the head end **38**, i.e. moved to a Trendelenburg orientation, controller **76** adjusts the boundary **102** of the alert zone **98** by moving at least one point of the boundary **102** closer to the head end **38**. The boundary **102** can have its longitudinal length increased in the direction of head end **38** to increase the area covered by the alert zone **98**, or the entire boundary **102** can be moved toward the head end **38** to relocate the alert zone **98** closer to the head end **38** without changing the overall size of the zone **98**. Either option allows the occupant's center of gravity **96** to approach the head end **38** of the person support apparatus **20** more closely than it otherwise could (if the alert zone **98** remained the same as shown in FIG. **18**) without triggering an exit alert.

(103) Controller **76** also adjusts the boundary of the arming zone **100** by moving at least one portion of the boundary **104** closer to the head end **38**. The boundary **104** can have its longitudinal length increased in the direction of head end **38** to increase the area covered by the arming zone **100**, or the entire boundary **104** can be moved toward the head end **38** to relocate the arming zone **100** closer to the head end **38** without changing the overall size of the zone **100**. Either option allows the occupant's center of gravity **96** to be closer to the head end **38** than it otherwise could (if the arming zone **100** remained the same as shown in FIG. **18**) while still permitting arming of the exit detection system **58**.

(104) As can be seen in FIG. **20**, when the litter frame **28** is inclined to elevate the head end **38** above the foot end **40**, i.e. moved to a reverse Trendelenburg orientation, controller **76** adjusts the boundary **102** of the alert zone **98** by moving at least one point of the boundary **102** closer to the foot end **40**. The boundary **102** can have its longitudinal length increased in the direction of foot end **40** to increase the area covered by the alert zone **98**, or the entire boundary **102** can be moved toward the foot end **40** to relocate the alert zone **98** closer to the foot end **40** without changing the overall size of the zone **98**. Either option allows the occupant's center of gravity **96** to approach the foot end **40** of the person support apparatus **20** more closely than it otherwise could (if the alert zone **98** remained the same as shown in FIG. **18**) without triggering an exit alert.

(105) Controller **76** also adjusts the boundary of the arming zone **100** by moving at least one portion of the boundary **104** closer to the foot end **40**. The boundary **104** can have its longitudinal length increased in the direction of foot end **40** to increase the area covered by the arming zone **100**, or the entire boundary **104** can be moved toward the foot end **40** to relocate the arming zone **100** closer to the foot end **40** without changing the overall size of the zone **100**. Either option allows the occupant's center of gravity **96** to be closer to the foot end **40** than it otherwise could (if the arming zone **100** remained the same as shown in FIG. **18**) while still permitting arming of the exit detection system **58**.

(106) With respect to the incline of the person support apparatus **20**, exit detection system **58** receives this information either directly from angle sensors **78b**, position sensors **78c**, or it is input into exit detection system **58**. When input, it may be input via a caregiver directly entering the incline angle information via user interface **82**, or it may be input by a caregiver selecting a position control, for example a one-touch input control to move the person support apparatus **20** to the Trendelenburg orientation or reverse Trendelenburg orientation. Regardless of the source of the

angle data, controller **76** uses this information to adjust one or more of the zones **98, 100**. That is, controller **76** uses the angle to adjust one or more of the boundaries **102, 104** of one or more of the zones **98, 100** based upon this angle.

(107) In one embodiment, controller **76** adjusts the location of the boundary **102, 104** of a zone **98, 100** based upon the current incline angle of person support apparatus **20**. Adjusting the location of the boundary **102, 104** as the person support apparatus **20** is tilted will allow the occupant's center of gravity **96** to move closer to the head end **38** or foot end **40** of person support apparatus **20** without triggering an exit alert, or it will prevent the occupant from moving as close to the ends **38, 40** of the person support apparatus **20** before triggering an exit alert.

(108) In some embodiments, controller **76** moves a zone **98, 100** to a predetermined location in response to movement of the person support apparatus **20** to the Trendelenburg orientation or reverse Trendelenburg orientation. That is, the zone **98, 100** is moved to a first location when the person support apparatus **20** is in the Trendelenburg orientation and the zone **98, 100** is moved to a second location when the person support apparatus **20** is in the reverse Trendelenburg orientation.

(109) Controller **76** is also programmed in some embodiments to change the shape of one or more zones **98, 100** based upon the incline angle of litter frame **28**. For example, the boundary **102, 104** of the zone **98, 100** can be widened near the head end **38** when the person support apparatus **20** is in the Trendelenburg orientation and the boundary **102, 104** of the zone **98, 100** can be widened near the foot end **40** when the person support apparatus **20** is in the reverse Trendelenburg orientation.

(110) Although FIGS. **18-20** illustrate only a single alert zone **98** and a single arming zone **100**, it will be understood that, in some embodiments, controller **76** changes the size, shape, and/or location of multiple zones in response to the changing incline of the litter frame **28**. Further, it will be understood that the adjustment in size of the zones **98, 100** illustrated in FIGS. **18-20** are merely one example of the types of zone adjustments that may be implemented by controller **76** in response to the changing incline angle. Other types of zone adjustments are also possible, including an adjustment in location and/or shape.

(111) In at least one embodiment, the distance that controller **76** shifts the zone **98** toward the head end **38** or foot end **40** in response to changes in the incline of litter frame **28** is determined based the distance the patient's measured center of gravity changes due solely to the change in incline angle. In other words, controller **76** is configured to use a known trigonometric ratio to calculate the distance the center of gravity of the patient moves based solely upon changes in the incline of the litter frame **28**, and to use this distance to adjust the length or position of zone **98** toward head end **38** or foot end **40**.

(112) FIGS. **21-22** illustrate one manner in which controller **76** adjusts the boundary of multiple user-selectable zones in response to the width being changed. Taking just the alert zones as an example, as can be seen in FIG. **21**, when the litter frame **28** is inclined to elevate the foot end **40** above the head end **38**, i.e. moved to a Trendelenburg orientation, controller **76** adjusts the boundaries **102a, 102b, 102c** of alert zones **98a, 98b, 98c** by moving at least a portion of the boundaries **102a, 102b, 102c** closer to the head end **38**. The boundary of their associated arming zones (not shown in FIG. **21**) is similarly adjusted.

(113) As can be seen in FIG. **22**, when the litter frame **28** is inclined to elevate the head end **38** above the foot end **40**, i.e. moved to a reverse Trendelenburg orientation, controller **76** adjusts the boundaries **102a, 102b, 102c** of alert zones **98a, 98b, 98c** by moving at least a portion of the boundaries **102a, 102b, 102c** closer to the foot end **40**. The boundary of their associated arming zones (not shown in FIG. **22**) is similarly adjusted.

(114) In the embodiment shown in FIGS. **21** and **22**, all of the user-selectable zones are adjusted in response to the incline angle being changed. In other embodiments, one or more zones may be static with respect to incline angle and not have their boundaries adjusted when the litter frame **28** is tilted.

(115) Controller **76** is also programmed in some embodiments to take into account the status of a lateral rotation therapy of the person support apparatus **20** when adjusting the boundary **102, 104** of the zones **98, 100**. In lateral rotation therapy, sometimes referred to simply as rotation therapy, a patient supported on a mattress is rotated laterally in an effort to reduce pulmonary complications of immobility. Controller **76** receives output signals from a mattress **140** via a mattress interface, and those output signals inform controller **76** of the status of a lateral rotation therapy.

Alternatively, controller **76** may receive the status information of the lateral rotation therapy from user interface **82**. Regardless of the source of this information, controller **76** uses the status information to adjust the boundary **102, 104** of one or more of the zones **98, 100**. Some non-limiting examples of lateral rotation therapy status include: a direction in which the occupant is turned; a lateral inclination angle; or pressure in an air bladder of the mattress.

(116) FIG. **23** is a transverse cross-section of a mattress **140** that can be disposed on the support deck **30** (FIG. **1**). The mattress **140** provides a patient support surface **142** upon which an occupant **O** is supported. As illustrated, the mattress **140** includes cushioning material **144** within a cover **146** and the cushioning material **144** may be a conventional bedding material, such as, but not limited to foam, polymeric materials, gels, or combinations thereof. A main air bladder **148** is positioned within mattress **140** and immediately below an upper portion of the cover **146**. The main air bladder **148** acts as the primary support for the patient.

(117) A rotation device **150** is positioned below the main air bladder **148**. The rotation device **150** provides lateral rotation therapy to the patient by rotating the patient from side to side. Rotation therapy assists in reducing bed sores and pulmonary problems of the patient. The rotation device **150** includes a pair of longitudinally positioned rotation bladders, including a first selectively inflatable and deflatable air bladder **152L** on a first lateral side (i.e. left side) of the mattress **140** and a second selectively inflatable and deflatable air bladder **152R** on a second lateral side (i.e. right side) of the mattress **140**. FIG. **24** illustrates lateral rotation of the occupant **O** to about a 15 degree angle to left side by inflating the right side bladder **152R**, and optionally deflating the left side bladder **152L**. FIG. **25** illustrates lateral rotation of the occupant **O** to about a 15 degree angle to right side by inflating the left side bladder **152L**, and optionally deflating the right side bladder **152R**.

(118) For static lateral rotation therapy, at least one of the rotation bladders **152L, 152R** is inflated to raise one side of the occupant **O**. For continuous lateral rotation therapy, the rotation bladders **152L, 152R** are inflated and deflated in sequential alternation to raise one side of the occupant **O**, lower the occupant **O**, and then raise the other side of the occupant **O** such that the occupant **O** experiences a side-to-side rotation that shifts pressures between the occupant **O** and the mattress **140**.

(119) The mattress **140** can further be configured to provide other conventional mattress functions, including turn assist, alternating pressure therapy, percussion therapy, vibration therapy, and low airloss therapy. The particular structural details of mattress **140** can vary widely. One exemplary mattress **140** that may be used with the person support apparatus **20** is described in U.S. Pat. No. 8,413,271 to Blanchard et al., entitled PATIENT SUPPORT APPARATUS, the complete disclosure of which is incorporated herein by reference.

(120) Referring additionally to FIG. **5**, the mattress **140** includes a mattress controller **154** used in controlling one or more functions of the mattress **140**, including lateral rotation therapy, and the independent inflation and deflation of the rotation bladders **152L, 152R**. A mattress interface **156** of the person support apparatus **20** receives status data from the mattress **140**. Such information may come from the mattress controller **154** and/or user interface **82**. The mattress interface **156** can be any suitable communication interface for communication between mattress **140** and the controller **76**, or between user interface **82** and controller **76**. In some embodiments, mattress interface **156** is a transceiver for an embedded network on person support apparatus **20**, such as, but not limited to, a Controller Area Network (CAN) transceiver, which carries status information communicated

between controller **76** and mattress controller **154** (and user interface **82**). In other embodiments, mattress interface **156** may be a LVDS (Low-Voltage Differential Signaling) line coupled to controller **76**, an I-squared-C connection, or still another type of communication line or bus.

(121) Via the user interface **82**, a caregiver can navigate to a rotation therapy screen **160**, shown in FIG. **29**, which is displayed on the display **86** of the user interface **82**. Therapy screen **160** includes one or more controls for inputting a desired rotation therapy program, including inputting desired patient orientation(s) and the hold time per orientation. Specifically, the screen **160** includes at least one hold control for selecting how long the patient is held in a particular orientation and at least one direction control for selecting a direction in which the patient is rotated.

(122) Therapy screen **160** also includes multiple hold controls **162**, **164**, **168** for respectively selecting a hold time for rightward incline of the patient, leftward incline of the patient, and a non-rotated or horizontal orientation of the patient between turns. Upon selection of a zero ("0") hold time for any of the orientations, any of the leftward incline, rightward incline, or horizontal positions are eliminated the therapy program.

(123) As shown in FIG. **29**, therapy screen **160** also includes multiple direction controls **170**, **172** for respectively selecting the degree of rightward and leftward rotation, i.e. the lateral inclination angle. By touching and sliding controls **170**, **172** along the length of the illustrated arc, the caregiver is able to select the specific amount of rotation he or she would like the mattress **140** to implement during each turn of the lateral rotation therapy.

(124) A start control **174** is provided on therapy screen **160** for initiating lateral rotation therapy once a desired therapy program is set. Based on the settings of the currently selected rotation program, mattress controller **154** (FIG. **5**) inflates and/or deflates one of more of the bladders **152L**, **152R** as needed to carry out the selected rotation program including controlling the pressure in one or more of the bladders **152L**, **152R**.

(125) The controller **76** is in communication with the mattress interface **156** and is configured to receive status data from mattress **140** via the mattress interface **156**. From this status data, the controller **76** determines a status of the lateral rotation therapy and adjusts the boundary **102**, **104** of one or more zones **98**, **100** based on the status of the lateral rotation therapy. Some of the factors or criteria used by controller **76** to vary the boundary **102**, **104** of one or more zones **98**, **100** include: the direction in which the occupant **O** is turned (i.e. rightward rotation or rotation toward a right side of the mattress **140** vs. leftward rotation or rotation toward a left side of the mattress **140**), the lateral inclination angle (i.e. the degree to which the occupant **O** is rotated rightward or leftward), or pressure in one or more of the rotation bladders **152L**, **152R**. Such information may be determined by one or more sensors of the mattress **140**.

(126) FIGS. **23-38** illustrate one manner in which controller **76** adjusts the boundary **102**, **104** of the zones **98**, **100** in response to the status of lateral rotation therapy. FIG. **26** illustrates the alert zone **98** and arming zone **100** for the person support apparatus **20** when the mattress **140** in a non-rotated or horizontal orientation as shown in FIG. **23**. When the occupant **O** of the mattress **140** moves such that his or her center of gravity **96** travels outside of alert zone **98**, i.e. crosses the boundary **102**, controller **76** issues an alert. When the center of gravity **96** of the occupant **O** is inside of arming zone **100**, i.e. within the boundary **104**, controller **76** may arm the exit detection system **58**.

(127) When the mattress **140** is at a leftward incline as shown in FIG. **24**, controller **76** adjusts the boundary **102** of the alert zone **98** by moving at least one point of the boundary **102** leftward as shown in FIG. **27**. The boundary **102** can be moved outward along at least one side of the alert zone **98** corresponding to the leftward side of the mattress **140** to increase the area covered by the alert zone **98**. Alternatively, the entire boundary **102** can be moved leftward to relocate the alert zone **98** closer to the leftward side of the mattress **140** without changing the overall size of the zone **98**. Either option allows the occupant's center of gravity **96** to approach the leftward side of the mattress **140** more closely than it otherwise could (if the alert zone **98** remained the same as shown

in FIG. 26) without triggering an exit alert.

(128) Controller **76** also adjusts the boundary of the arming zone **100** by moving at least one portion of the boundary **104** leftward. The boundary **104** can be moved outward along at least one side of the arming zone **100** corresponding to the leftward side of the mattress **140** to increase the area covered by the arming zone **100**. Alternatively, the entire boundary **104** can be moved leftward to relocate the arming zone **100** closer to the leftward side of the mattress **140** without changing the overall size of the zone **100**. Either option allows the occupant's center of gravity **96** to be closer to the leftward side of the mattress **140** than it otherwise could (if the arming zone **100** remained the same as shown in FIG. 26) while still permitting arming of the exit detection system **58**.

(129) When the mattress **140** is at a rightward incline as shown in FIG. 25, controller **76** adjusts the boundary **102** of the alert zone **98** by moving at least one point of the boundary **102** rightward as shown in FIG. 28. The boundary **102** can be moved outward along at least one side of the alert zone **98** corresponding to the rightward side of the mattress **140** to increase the area covered by the alert zone **98**. Alternatively, the entire boundary **102** can be moved rightward to relocate the alert zone **98** closer to the rightward side of the mattress **140** without changing the overall size of the zone **98**. Either option allows the occupant's center of gravity **96** to approach the rightward side of the mattress **140** more closely than it otherwise could (if the alert zone **98** remained the same as shown in FIG. 26 or in FIG. 27) without triggering an exit alert.

(130) Controller **76** also adjusts the boundary of the arming zone **100** by moving at least one point of the boundary **104** rightward. The boundary **104** can be moved outward along at least one side of the arming zone **100** corresponding to the rightward side of the mattress **140** to increase the area covered by the arming zone **100**. Alternatively, the entire boundary **104** can be moved rightward to relocate the arming zone **100** closer to the rightward side of the mattress **140** without changing the overall size of the zone **100**. Either option allows the occupant's center of gravity **96** to be closer to the rightward side of the mattress **140** than it otherwise could (if the arming zone **100** remained the same as shown in FIG. 26 or in FIG. 27) while still permitting arming of the exit detection system **58**.

(131) In some embodiments, controller **76** dynamically adjusts the zones **98**, **100** based upon a dynamically changing rotation direction, i.e. during continuous lateral rotation therapy. In such embodiments, controller **76** can shift the zone **98**, **100** laterally back and forth as the occupant **O** experiences side-to-side rotation.

(132) With respect to the status of the lateral rotation therapy, exit detection system **58** may receive at least some of this information from mattress interface **156**. Exit detection system **58** may also determine the status of the lateral rotation therapy based at least in part on information input into exit detection system **58** via the user interface **82**. When input, it may be input via a caregiver entering hold times or rotation directions via the input controls **162-172** on the therapy screen **160** (FIG. 29). In one embodiment, upon selection of the start control **174**, the controller **76** can initiate a zone adjustment based on the status of lateral rotation therapy. Regardless of the source of the status data, controller **76** uses this information to adjust one or more of the zones **98**, **100**. That is, controller **76** uses the status of lateral rotation therapy to adjust one or more of the boundaries **102**, **104** of one or more of the zones **98**, **100** based upon this status.

(133) Controller **76** is also programmed, in some embodiments, to take into account a non-occupant object added to or removed from the support deck **30** when adjusting the boundary **102**, **104** of the zones **98**, **100**. Controller **76** can determine or detect the addition or removal of a non-occupant object based on the output signals of the force sensors **56** and/or based on input from a caregiver, such as by a caregiver logging the addition or removal of an object via user interface **82**. Regardless of the source of this information, controller **76** uses the information to adjust the boundary **102**, **104** of one or more of the zones **98**, **100**.

(134) FIGS. 30-32 illustrate one manner in which controller **76** adjusts the boundary **102**, **104** of the zones **98**, **100** based on the addition of a non-occupant object on the support deck **30**.

Specifically, controller **76** is configured to adjust the boundary **102, 104** according to a change in the center of gravity after the addition of a non-occupant object. In the illustrative example shown in FIG. **30**, controller **76** has determined the center of gravity of the mass on the support surface to be at a location **96A**, the mass comprising the mass of an occupant (not shown). After determining that a non-occupant object **180** has been added to the support surface as shown in FIG. **31**, the controller **76** compares the center of gravity **96A** immediately prior to the addition event to a current center of gravity **96B**, i.e. the center of gravity immediately after the addition event. During the comparison of the calculated centers of gravity **96A, 96B**, controller **76** determines which direction the center of gravity has moved, and the magnitude of the change in the center of gravity. The direction and magnitude are represented in FIG. **31** by vector **182**. This direction and magnitude are used by controller **76** in its adjustment of one or more of the zones **98, 100**. For example, if the center of gravity shifts leftward and toward the head end **38** of person support apparatus **20** as shown in FIG. **31**, controller **76** will use this information to shift one or more of the zones **98, 100** leftward and toward the head end **38** of person support apparatus **20** as shown in FIG. **32**. More specifically, each point of the boundary **102, 104** of one or more of the zones **98, 100** are shifted in the same direction and by the same magnitude as represented by vector **182**.

(135) FIGS. **33-35** illustrate one manner in which controller **76** adjusts the boundary **102, 104** of the zones **98, 100** based on the removal of a non-occupant object from the support deck **30**. Specifically, controller **76** is configured to adjust the boundary **102, 104** according to a change in the center of gravity after the removal of a non-occupant object. In the illustrative example shown in FIG. **33**, controller **76** has determined the center of gravity of the mass on the support surface to be at a location **96A**, the mass comprising the mass of an occupant (not shown) and the mass of non-occupant object **180**. After determining that non-occupant object **180** has been removed from the support surface as shown in FIG. **34**, the controller **76** compares the center of gravity **96A** immediately prior to the removal event to a current center of gravity **96B**, i.e. the center of gravity immediately after the removal event. During the comparison of the calculated centers of gravity **96A, 96B**, controller **76** determines which direction the center of gravity has moved, and the magnitude of the change in the center of gravity. The direction and magnitude are represented in FIG. **34** by vector **184**. This direction and magnitude are used by controller **76** in its adjustment of one or more of the zones **98, 100**. For example, if the center of gravity shifts rightward as shown in FIG. **34**, controller **76** will use this information to shift one or more of the zones **98, 100** rightward as shown in FIG. **35**. More specifically, each point of the boundary **102, 104** of one or more of the zones **98, 100** are shifted in the same direction and by the same magnitude as represented by vector **184**.

(136) Although FIGS. **30-35** illustrate only a single alert zone **98** and a single arming zone **100**, it will be understood that controller **76** adjusts all zones **98, 100** in response to the addition or removal of a non-occupant object, in at least some embodiments. Further, it will be understood that the zone changes illustrated in FIGS. **30-35** are merely one example of the types of changes that may be implemented by controller **76** in response to the addition or removal of a non-occupant weight. Other types of changes are also possible. One example is shown in FIG. **36**, where the controller **76** changes the size and shape of the boundary **102, 104** from that shown in FIG. **30** based on the addition of a non-occupant object on the support deck **30**. After determining that a non-occupant object **180** has been added to the support surface near one of the force sensors, designated **56A** in FIG. **36**, controller **76** will use this information to expand one or more of the zones **98, 100** toward the force sensor **56A**.

(137) As an example of another type of change implemented by controller **76** in response to the addition or removal of an object, in at least one embodiment, controller **76** is configured to reduce the size of the medium and large alerts zones **98b** and **98c** whenever an object is added, but leave the size of small alert zone **98a** unchanged. Additionally, the degree to which the medium and large alert zones **98b** and **98c** are adjusted is based upon the weight of the added or removed object (the

heavier the object that is added, the smaller controller **76** shrinks the zones **98b** and **98c**; and the heavier the object, the greater controller **76** expands the zones **98b** and **98c** when the object is removed).

(138) In yet another embodiment, controller **76** is configured to not make any adjustments to the alert zones **98a-c** in response to the addition or removal of an object, but is instead adapted to adjust the readings from force sensors **56** such that the effect of the object's removal or addition is removed from controller **76**'s subsequent calculations of the center of gravity. In this embodiment, controller **76** records the outputs from each of the four sensors **56** repetitively (e.g. multiple times a second) and, after an object is added or removed, it compares each of the readings from the individual force sensors **56** with the reading from that same force sensor immediately prior to the addition or removal of the object. From this comparison, controller **76** determines the difference in each force sensor **56**'s readings between the moment immediately before the object was added (or removed) and immediately after the object was added (or removed). This difference is stored in memory and controller **76** thereafter adjusts the outputs from each individual force sensor **56** by its corresponding difference before computing the center of gravity used for triggering the exit alert.

(139) For example, suppose an eleven pound object is added to the support deck **30**, and, due to the particular position on support deck at which the object is placed, the result is that a first force sensor **56a** registers a two pound increase in weight, a second force sensor **56b** registers a two and a half pound increase in weight, a third force sensor **56c** registers a three pound increase in weight, and the fourth force sensor **56d** registers a three and a half pound increase in weight. Controller **76** computes each of these changes in weight and stores not only the values of these differences, but also which force sensor **56** each of the differences corresponds to. In other words, in this example, controller **76** stores information identifying the first force sensor **56a** weight change as plus two pounds; information identifying the second force sensor **56b** weight change as plus two and a half pounds; information identifying the third force sensor **56c** weight change as plus three pounds; and information identifying the fourth force sensor **56d** weight change as plus three and a half pounds. After storing this information in memory, controller **76** thereafter uses these values to adjust the outputs of each of the force sensors **56** before calculating the occupant's center of gravity to determine whether it should issue an exit alert or not. Thus, in this particular example, after the eleven pound object is added, controller **76** subsequently subtracts two pounds from each force reading from first force sensor **56a**, subtracts two and a half pounds from each force reading from second force sensor **56b**, subtracts three pounds from each force reading from third force sensor **56c**, and subtracts three and a half pounds from each force reading from fourth force sensor **56d**. After subtracting these values, controller uses the reduced values from the four sensors **56a-d** to compute the occupant's center of gravity, which it then compares with the boundary **102** of the currently active alert zone **98**. By making these subtractions, the effect of the object's weight on the calculated center of gravity (assuming the object remains stationary) is effectively zeroed out.

(140) If an additional object is added, controller **76** uses the same procedure to zero out the object's influence on the center of gravity calculations of the occupant. That is, it determines the value of the added weight on each individual force sensor **56**, stores those value, and then subtracts those values (along with the values from the first added object) from the corresponding force sensors **56** before calculating the occupant's center of gravity. If an object is removed from support deck **30**, controller **76** simply stops performing these subtractions.

(141) It should be noted that this object zeroing out is separate from, and in addition to, the overall zeroing of the scale system of person support apparatus **20** that uses force sensors **56** to detect the occupant's weight. In order to accurately determine the occupant's weight, the weight that is sensed by force sensors **56** due to the weight of litter frame **28**, deck **30**, siderails **36**, mattress **140**, any bedding and/or pillows on mattress **140**, etc., must be subtracted from the total weight readings from force sensors **56**. In order to accomplish this, the user of person support apparatus **20** executes a zeroing (or taring) process while the occupant is positioned off of person support apparatus **20**.

This zeroing process accounts for non-patient weight that is present before the patient enters the person support apparatus **20**. Thus, this zeroing process is different from the above-described object-based zeroing process that is executed by controller **76** after the occupant is positioned on support deck **30** and that is used to make adjustments to the center of gravity in order to zero out the effect of added or removed objects on the occupant's calculated center of gravity.

(142) In some embodiments, controller **76** is configured to zero out the effects of added or removed objects on the occupant's center of gravity at all times, regardless of whether or not exit detection system **58** is armed. Thus, for example, if exit detection system **58** is not armed, but an object is added, controller **76** detects and records the increased weight values for each force sensor **56** and, if exit detection system **58** is subsequently armed, uses those weight values to zero out the effect of the added object's weight on the occupant's center of gravity calculations. In at least one of these embodiments, controller **76** is configured to use these pre-arming object additions or object removals (i.e. objects added or removed prior to exit detection system **58** being armed) only for zeroing out the object's weight when the user subsequently arms exit detection system **58** and selects one of zones **98b** or **98c** as the active zone. If the user subsequently arms exit detection system **58** and selects zone **98a** as the active zone, controller **76** does not zero out the weight of the pre-arming object addition or object removal. This is because, as noted previously, in at least one embodiment, controller **76** is configured to define the location of the smallest zone **98a** based on the location of the occupant's center of gravity at the moment the exit detection system is armed, while controller **76** is configured to define the location of the medium and largest zones **98b** and **98c** at locations that are fixed with respect to frame of reference **88** (i.e. locations that are independent of the patient's location when the exit detection system **98** is armed).

(143) Exit detection system **58** determines whether an object has been added or removed either automatically based on output signals from the force sensors **56**, or in response to information input by a user via user interface **82**. In one embodiment, the controller **76** is configured to automatically detect the addition or removal of a non-occupant object. The controller **76** monitors the signals from the plurality of force sensors **56** and analyzes the signals to determine if the signals are indicative of a non-occupant object moving onto or off the support surface. More specifically, controller **76** examines the received signals to determine if the signals represent a change in weight greater than a weight threshold that occurs during a time period less than a time threshold. If the weight change is less than the threshold or if the change occurs gradually, then controller **76** interprets these changes as being due to the patient's weight slowly changing. If the weight change is at or above the threshold or if weight change occurs suddenly, then controller **76** interprets this as an object being added to or removed from the support deck **30**. Methods for automatically identifying the addition or removal of a non-patient object on a person support apparatus using force sensors **56** are disclosed in commonly assigned U.S. Patent Application Publication No. 2016/0022218 to Hayes et al., entitled PATIENT SUPPORT APPARATUS WITH PATIENT INFORMATION SENSORS, the complete disclosure of which is hereby incorporated herein by reference. Once a weight change identified as being due to the non-occupant object, controller **76** can adjust the boundary **102**, **104** as described above with respect to FIGS. **30-36**, or alternatively zero-out the effect of the object's added or removed weight on the occupant's center of gravity calculations such that the adjustments to the boundaries **102**, **104** can be omitted.

(144) In another embodiment, controller **76** receives user input that a non-occupant object has been added to or removed from the person support apparatus **20**. This input may be communicated to controller **76** via a caregiver entering the information into user interface **82**. A system for logging the addition and removal of equipment or other non-occupant objects for a person support apparatus is disclosed in commonly assigned U.S. patent application Ser. No. 62/885,954, filed Aug. 13, 2019, by inventors Kurosh Nahavandi et al. and entitled PATIENT SUPPORT APPARATUS WITH EQUIPMENT WEIGHT LOG, the complete disclosure of which is hereby incorporated herein by reference. Once a weight change is identified as being due to a non-

occupant object, controller **76** adjusts the boundary **102**, **104** as described above with respect to FIGS. **30-36**.

(145) In those embodiments where exit detection system **58** is configured to automatically detect the addition or removal of a non-occupant object, controller **76** may additionally be configured to attempt to automatically identify the added or removed object by comparing the detected weight change to a database. The database contains a listing of weight change ranges that are mapped to potential objects or reasons for the weight change. For example, the database may characterize any weight changes of 100 pounds or greater as likely corresponding to a patient entering or exiting the person support apparatus **20**. Other thresholds can, of course, be used (particularly if the person support apparatus **20** is used in a pediatric setting). The database might also characterize any weight changes of less than 100 pounds but greater than the weigh threshold as likely corresponding to a non-occupant object being added or removed. Changes in weight greater than this threshold, but less than the threshold corresponding to a patient entering or exiting, could be assigned as corresponding to medical equipment being removed or added. For greater accuracy, the weights of medical devices, bedding, and/or other objects that are commonly used in conjunction with the person support apparatus **20** can be input into the database. Methods for automatically distinguishing between patient weight and non-patient objects on a person support apparatus are disclosed in commonly assigned U.S. Patent Application Publication No. 2016/0022218 to Hayes et al., incorporated above.

(146) Controller **76** is also programmed in some embodiments to take into account other inputs when adjusting the size, shape, and/or location of one or more zones **98**, **100**. For example, the controller **76** can take into account any one or more of: the length of the deck (in those patient supports having a deck that is adjustable in the longitudinal direction), the gender of the occupant, the race or ethnicity of the occupant, an assessment of the fall risk of the occupant, readings of one or more vital signs of the occupant, the state of the siderails **36**, an amount of downward force that is being applied to one or more of the siderails **36** (i.e. based on input from siderail force sensors **78**), the position of one or more other components of person support apparatus **20** (i.e. based on input from position sensors **78c**), one or more sequences of movement of the occupant of person support apparatus **20**, the environment of person support apparatus **20**, the proximity of a caregiver to person support apparatus **20**, the current time, the amount of time that has passed since the occupant last exited from person support apparatus **20**, additional information regarding the medical condition of the occupant, an orientation of the occupant's body, one or more outputs from an electromyograph (EMG) when adjusting the size of one or more zones **98**, **100**. One example of the manner in which controller **76** may be programmed to such factors is disclosed in commonly assigned U.S. Patent Application Publication No. 2017/0098359 to Sidhu et al., incorporated above.

(147) As mentioned previously, the foregoing examples of the factors that may be used by exit detection system **58** are not necessarily all used in combination with each other. Instead, in some embodiments, exit detection system **58** uses only one of these factors to adjust one or more of the zones **98**, **100**, while in other embodiments, exit detection system **58** uses a combination of these factors to adjust one or more zones **98**, **100**. When using multiple factors to adjust exit detection system **58**, the different factors may cause changes to the exit detection system at different times and be cumulative to each other. For example, when exit detection system **58** takes into account the width of the support deck **30**, controller **76** changes an initial size or shape of at least one zone **98**, **100** and subsequently uses the adjusted size or shape of that zone as a base zone. Thereafter, further adjustments to the base zone may be made, depending upon the particular factors that controller **76** has been programmed to take into account. For example, if the litter frame **28** is inclined, controller **76** then makes an additional adjustment to the base zone. Similarly, if a non-occupant object is added to or removed from the person support apparatus **20**, controller **76** may then make yet another adjustment in addition to the ones previously made. Exit detection system **58** therefore

dynamically responds to one or more changing conditions when determining whether to adjust one or more of the zones **98**, **100**.

(148) In at least one embodiment, person support apparatus **20** and controller **76** are configured to adjust the smallest zone **98a** in the manners set forth below in table 1 for any one or more of the factors listed therein:

(149) TABLE-US-00001

TABLE 1	Smallest Zone	Monitored	Monitored	Adjust	Adjust	Adjust																													
While System	While System	Zone	Zone	Zone	Factor	Disarmed?	Armed?	Shape?	Size?	Location?																									
Deck Width	No	No	No	No	Deck Length	No	No	No	No	Litter Frame Incline Angle	No	Yes																							
No	No	Yes	Litter Frame Height	No	No	No	No	No	Lateral Rotation	Therapy	No	Yes																							
No	No	Yes	Litter Frame Height	No	No	No	No	No	Lateral Rotation	Therapy	No	Yes																							
Object Addition/Removal	No	Yes	No	No	Yes	Siderail Position	No	Yes	No	No	Yes	Head Section Orientation	No	Yes	No	No	Yes	Occupant Characteristics	N/A	N/A	No	No	No	(gender, height, weight, fall risk, vital signs, medical condition)	Time of Day	No	No	No	No	Caregiver Proximity	No	No	No	No	No

(150) In this particular embodiment, person support apparatus **20** is also configured to adjust the medium and large zones **98b** and **98c** in the manners set forth below in table 2 for any one or more of the factors listed therein:

(151) TABLE-US-00002

TABLE 2	Medium and Large Zones	98b	and	98c	Monitored	Monitored	Adjust	Adjust	while System	While System	Zone	Adjust	Zone	Factor	Disarmed?	Armed?	Shape?	Zone Size?	Location?	Deck Width	No	Yes	Yes	Yes	No	Deck Length	No	Yes	Yes	Yes	Yes	Litter Frame Incline Angle	No	Yes	Yes	Yes	Yes	Litter Frame Height	No	Yes	No	Yes	No	Lateral Rotation	Therapy	No	Yes	No	No	Yes	Object Addition/Removal	Yes	Yes	No	No	Yes	Siderail Position	No	Yes	Yes	Yes	No	Head Section Orientation	No	Yes	Yes	Yes	No	Thigh/Foot Section Orientation	No	Yes	Yes	Yes	No	Occupant Characteristics	N/A	N/A	Yes	Yes	No	(gender, height, weight, fall risk, vital signs, medical condition)	Time of Day	No	Yes	No	Yes	No	Caregiver Proximity	No	Yes <th>Yes</th> <th>Yes</th> <th>Yes</th> <td>No</td>	Yes	Yes	Yes	No
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(152) The second column of these tables indicates whether the corresponding factor is monitored by controller **76** when the exit detection system **58** is disarmed. The third column indicates whether the corresponding factor is monitored by controller **76** when the exit detection system **58** is armed. The fourth column indicates whether controller **76** uses that factor to make adjustments to the shape of the corresponding zone(s) **98** when the exit detection system **58** is armed. Finally, the fifth and sixth columns indicate whether controller **76** uses that factor to make adjustments to the size and location, respectively, of the corresponding zone(s) **98**. It will, of course, be understood that, many modifications can be made to these tables.

(153) From these two tables, it can be seen that controller **76** adjusts zone **98a** in a different manner from zones **98b** and **98c**. For example, controller **76** is adapted to monitor the addition and removal of objects at all times (regardless of whether exit detection system **58** is armed or not) and use those additions and removals to adjust zones **98b** and **98c**, but not **98a**. It can also be seen from the fourth through sixth columns that controller **76** does not make any types of adjustments to zone **98a** for several factors when the exit detection system **58** is armed (e.g. deck width, litter height, time of day, caregiver proximity), yet controller **76** does make one or more types of adjustments to zones **98b** and **98c** for these factors (and others) when exit detection system **58** is armed.

(154) Various modifications can be made to the embodiment illustrated in Tables 1 and 2. For example, in some embodiments, exit detection system **58** includes only two zones, and the exit detection system **58** adjusts a first one of the two zones based on one or more of the factors listed in Table 1 (and in the manner shown therein) and adjusts the second one of the two zones based on one or more of the factors listed in Table 2 (and in the manner shown therein). In another modified embodiment, exit detection system **58** includes three zones, but the medium and large zones **98b** and **98c** are adjusted independently for one or more of the factors listed in Table 2. In such embodiments, a third table (not shown) may be provided indicating the manners in which the third zone is adjusted independently of the manners illustrated in Table 2. Still other modifications are possible. In still another modified embodiment, exit detection system **58** is adapted to adjust the

smallest zone **98a** based on changes to the width of the deck and/or the length of the deck (for those patient support apparatuses having expandable widths and lengths, respectively).

(155) In some embodiments, controller **76** does not monitor one or more factors while the exit detection system **58** is disarmed, but makes changes to the zone(s) **98** immediately after exit detection system **58** is armed based on any differences between the current state of those factors and a baseline state of those factors as set forth in a predefined configuration stored in a memory **57** (FIG. 5) accessible to controller **76**. In other words, memory **57** stores a predefined location, size, and shape of one or more zones **98** that correspond to person support apparatus **20** being in the baseline state. For example, memory **57** might store a location, size, and shape of a zone **98b** corresponding to a support deck width X, a litter frame tilt angle of zero degrees, a litter frame height of Y, a raised position for all siderails **36**, a flat orientation for head section **42**, thigh section **46**, and foot section **48**, and one or more other conditions. When exit detection system **58** is initially armed, controller **76** compares the current states of these conditions to the baseline states set forth in memory **57** and, if they are different, makes immediate changes to the size, shape, and/or location of the zone **98** based on those differences. If additional changes to any of these conditions are subsequently made while exit detection system **58** is armed, controller **76** makes additional changes to the size, shape, and/or location of the zone **98**.

(156) Although exit detection system **58** has been primarily described herein as computing a center of gravity **96** of the occupant and comparing the position of the computed center of gravity to an active zone **98**, **100**, it will be understood by those skilled in the art that exit detection system **58** can be modified to process the outputs of force sensors **56** in other manners besides computing a center of gravity. For example, in some embodiments, controller **76** sums the total amount of force on force sensors **56** when person support apparatus **20** is occupied and then looks for shifts of more than a threshold amount of that weight to a side, head end, or foot end of person support apparatus. For example, if a 100 kilogram person is occupying person support apparatus **20**, exit detection system **58** may be modified to trigger an exit alert if more than X percent, say, 70% ($0.70 \times 100 = 70$ kilograms) of the total forces are detected by the two force sensors **56** positioned along the right side of person support apparatus **20**, or by the two force sensors **56** positioned along the left side of person support apparatus **20**. In some embodiments, a different ratio of the forces detected by the two force sensors **56** positioned along the foot end **40** of person support apparatus **20** may trigger an exit alert if the ratio exceeds a different threshold, while still another ratio of the forces detected by the two force sensors **56** positioned along the head end **38** of person support apparatus **20** may trigger an exit alert if that ratio exceeds yet a different threshold. In sum, exit detection system **58** can be modified to compute one or more ratios of the force detected by a first force sensor **56** (or the sum of forces detected by a combination of first force sensors **56**) to the force, or sum of forces, detected by at least one other force sensor **56**. The one or more ratios may then be compared to one or more thresholds for determining whether to issue an exit alert or not. Other types of weight distribution changes may also be used to trigger an exit alert.

(157) When exit detection system **58** is implemented to compute one or more force ratios based on the outputs of force sensors **56** instead of a center of gravity of the occupant, controller **76** modifies the threshold(s) used by exit detection system **58** in response to one or more of the factors discussed above. Thus, for example, exit detection system **58** may be programmed to issue an exit alert if 70% of the occupant's weight is detected on the right two force sensors **56** when both of the right siderails **36** are lowered, and to not issue an alert when both of the right siderails are raised until at least 80% of the occupant's weight is detected on the right two force sensors. Of course, these thresholds are merely illustrative, and different ones may be used.

(158) Further, exit detection system **58** may also be modified to use and analyze the outputs of non-force sensors, either in addition to or in lieu of the outputs from force sensors **56**. For example, the principles disclosed herein can be applied to a video image based exit detection system wherein an exit alert is issued if the position of the occupant meets one or more criteria (e.g. the occupant

moves to within X distance of a side of person support apparatus 20). Based on one or more of the factors discussed herein (e.g. width of the support deck 30, incline angle of the litter frame 28, lateral rotation therapy status, addition or removal of a non-occupant object, etc.), the exit detection system may alter one or more of the criteria (e.g. distance X) based upon these factors. Still other types of exit detection systems may be used in accordance with these principles, including, but not limited to, thermal imaging based exit detection systems, accelerometer based exit detection systems, radar based exit detection systems, pressure sensing exit detection systems, and others. (159) Various additional alterations and changes beyond those already mentioned herein can be made to the above-described embodiments. This disclosure is presented for illustrative purposes and should not be interpreted as an exhaustive description of all embodiments or to limit the scope of the claims to the specific elements illustrated or described in connection with these embodiments. For example, and without limitation, any individual element(s) of the described embodiments may be replaced by alternative elements that provide substantially similar functionality or otherwise provide adequate operation. This includes, for example, presently known alternative elements, such as those that might be currently known to one skilled in the art, and alternative elements that may be developed in the future, such as those that one skilled in the art might, upon development, recognize as an alternative. Any reference to claim elements in the singular, for example, using the articles “a,” “an,” “the” or “said,” is not to be construed as limiting the element to the singular.

Claims

1. A person support apparatus comprising: a support deck supported on a frame by at least one lift configured to raise and lower the support deck, the support deck having an adjustable width; and an exit detection system adapted to issue an alert if an occupant of the person support apparatus is moving toward exiting the person support apparatus, the exit detection system including an alert zone having a boundary, the exit detection system comprising: a plurality of force sensors adapted to output signals corresponding to downward forces exerted on the support deck; and a controller in communication with the plurality of force sensors, the controller configured to determine a deck width of the support deck and adjust the boundary of the alert zone based on the deck width.
2. The person support apparatus of claim 1 wherein the support deck comprises a plurality of expandable sections and a plurality of width sensors, each of the plurality of width sensors configured to detect the position of one of the plurality of the expandable sections, wherein the controller is in electrical communication with the plurality of width sensors and is configured to adjust the boundary in response to input from the plurality of width sensors.
3. The person support apparatus of claim 1 wherein the plurality of force sensors comprise a plurality of load cells adapted to detect a weight of the occupant, and wherein the controller is configured to calculate a center of gravity of the occupant from forces detected by the load cells and determine if the occupant is moving toward exiting the person support apparatus based upon whether the calculated center of gravity is inside or outside of the alert zone.
4. The person support apparatus of claim 1 wherein the controller is configured to determine a center of gravity of the occupant based on output signals from the plurality of force sensors and determine if the occupant is moving toward exiting the person support apparatus based upon whether the determined center of gravity is inside or outside of the alert zone.
5. The person support apparatus of claim 4 wherein the exit detection system comprises a plurality of user-selectable alert zones, wherein a first alert zone of the plurality of user-selectable alert zones has a first boundary dependent on the deck width and not dependent on the center of gravity of the occupant, wherein a second alert zone of the plurality of user-selectable alert zones has a second boundary that is dependent on the center of gravity of the occupant and not dependent on the deck width, and wherein a third alert zone of the plurality of user-selectable alert zones has a

third boundary dependent on the deck width and not dependent on the center of gravity of the occupant.

6. The person support apparatus of claim 1 wherein the controller is configured to reduce a width of the alert zone in response to the deck width being decreased and extend the width of the alert zone in response to the deck width being increased.

7. The person support apparatus of claim 1 wherein the exit detection system includes an arming zone having a boundary, wherein the controller is configured to adjust the boundary of the arming zone based on the deck width.

8. The person support apparatus of claim 1 wherein the boundary of the alert zone is adjustable by the controller based on the deck width and at least one additional criteria, and the at least one additional criteria comprises: an incline angle of the frame, a lateral rotation therapy status, the addition of a non-occupant object to the support deck, or the removal of a non-occupant object from the support deck.

9. The person support apparatus of claim 1 wherein the exit detection system further includes an arming zone having a boundary and the controller is further adapted to determine a condition of the person support apparatus; adjust the boundary of the arming zone based on the condition of the person support apparatus; and arm the exit detection system if the occupant is settled within the arming zone.

10. The person support apparatus of claim 9 wherein the plurality of force sensors comprise a plurality of load cells adapted to detect a weight of the occupant, and wherein the controller is configured to calculate a center of gravity of the occupant from forces detected by the load cells and determine if the occupant is settled within the arming zone based upon whether the calculated center of gravity is inside or outside of the arming zone.

11. The person support apparatus of claim 10, wherein the exit detection system comprises a plurality of user-selectable arming zones, wherein a first arming zone of the plurality of user-selectable arming zones has a first boundary dependent on the condition of the person support apparatus and not dependent on the center of gravity of the occupant, wherein a second arming zone of the plurality of user-selectable arming zones has a second boundary that is dependent on the center of gravity of the occupant and not dependent on the condition of the person support apparatus; and wherein a third arming zone of the plurality of user-selectable arming zones has a third boundary dependent on the condition of the person support apparatus and not dependent on the center of gravity of the occupant.

12. The person support apparatus of claim 10, wherein the exit detection system comprises a plurality of user-selectable arming zones, wherein a first arming zone of the plurality of user-selectable arming zones has a boundary dependent on the condition of the person support apparatus, and wherein a second arming zone of the plurality of user-selectable arming zones has a boundary that is not dependent on the condition of the person support apparatus.

13. The person support apparatus of claim 12, wherein the controller is configured to adjust the boundary of the arming zone by at least one of: changing a shape of the boundary, changing a dimension of the boundary, changing an area defined within the boundary, or changing a location of the boundary.

14. The person support apparatus of claim 12, wherein the condition of the person support apparatus comprises one of: a width of the support deck, an incline angle of the frame, a lateral rotation therapy status, the addition of a non-occupant object to the support deck, or the removal of a non-occupant object from the support deck.
